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Declaration

By virtue of this, I declare that this project is an original report of my research. It has been written by me and has not been submitted for any previous degree. The research work is entirely my own work and the affiliative contributions have been specified clearly and acknowledged. Any help taken on this thesis has been clearly mentioned in references. I have tried my best to provide all the information as my best and truest.

Gagandeep Kaur

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I hereby declare that this work is not confidential, and I gave Arden permission to save it a sample reference. I have tried my best to cover all the topics according to BTP Handbook of the Arden University.

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I also declare that I have already received and attached the Ethics certificate from my Module teacher in this Thesis.

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Abstract

Introduction: Dell Technologies is US based American company and has been one of the leading brands in the computer market with billions of revenues for over a decade now. For the past few years, the company's focus was on computers, servers, and other peripherals. But now the organization has decided to expand the business to new heights by launching the mobile phone in the market. The company has also tried previously the same business but was not successful and now has been ready with a comeback in the related market. Mobile industry is one of the leading industries now-a-days. Working on the loopholes and fixing them will lead this organization to new heights.

Strategies & Methods: As the company's focus is its customers, it clearly keeps in mind about daily needs and changes by them. Here we have clearly discussed the strategies this company uses and the industry trends. Here we have briefly explained the SWOT analysis, AGILE and 6 Sigma methodologies this company uses for the business.

Results & Findings: In this study, Dell has planned to launch the project of mobile phones with the possibility of an increase in the revenue for the company. Although there are risks in this business, the company is a well-known brand name and was single-handedly established by one single person, Mr. Michael Dell, and he is very ambitious to take the company to the next level.

Conclusions and Recommendations: The company needs to work a lot on this business project as there are so many competitors who are already pros in this field. The organization will have to work very aggressively in order to keep up with the changing demands of the customers.

Executive Summary

The following report includes Dell Technologies. (Henceforward referred to as Dell) products and services provided to consumers and commercials and the upcoming new projects of the organization with the possibility to increase the revenue for the company through various analysis. This project reports includes collecting information from secondary data using the company's website, blogs, user experience and reviews for the Dell products which will help the company in launching Dell mobile phones in the market. Technology changes very often. As computers and laptops are replaced by tablets, Dell needs to expand its product range. Dell has brought Dell Steak to market, but they need to improve their product line. In the new market, tablet PCs are also in great demand from customers such as iPad. Before the iPad's launch, analysts predicted it would be a major market failure, but the iPad proved to exceed expectations. Dell launched his Dell Steak but was supposed to release a tablet PC that could also support cell phone functionality. During the recession, US revenue fell 17% and Dell's shipments fell 23%. Because of the recession, organizations of all kinds have been greatly affected. As such, Dell was also indirectly involved in this recession. Dell's workforce reduction program is also part of the recovery from the recession, with Dell announcing a 12% reduction in workforce at its Irish site in 2008. Dell closed its manufacturing facility in Limerick, Ireland in January 2009 and moved the plant to Poland. With the incoming of covid-19, everything went to halt, and everyone became mobile as working from home Children's education became online and because of this, many people must buy smartphones as compared to laptops and desktops to continue studies and online classes. This also gave an idea to the organization to stay prepared for the future. [Dell, 2022](#); [Li, Y., \(2022\)](#)

Dell's business strategy is based on an effective supply chain model and quality manufacturing facilities. This business model provides customers with high quality, fast delivery and customized products. As Dell is a well-known brand in the electronics market, customers prefer to buy laptops and desktops over other brands, which has given the company a possibility and boost in the confidence of launching mobile phones again as well, which can take the company to further achievements and brand recognition among others around the world. Dell's products' prices are 15%-20% lower than other competitors in the market. Dell, in the past has already launched series of the below phone and now planning to launch new series:

Li, Y., (2022)

1. Dell Streak
2. Dell Venue
3. Dell Venue Pro

As in the recent news, dell has again planned to get into smartphone business. These first 3 mini smartphones will be first launched in China and Brazil with partnership with the biggest mobile operators, Claro and China Mobile as China has over 500 million subscribers, and Claro has over than 42 million users (about twice the population of New York). In addition, Dell has agreements with many other providers such as Vodafone in Europe, Australia, and New Zealand. This research will also include the methods used on this research which is qualitative approach and explained about the results and findings. DellTechnology., (2022)

CHAPTER 1. Introduction with Internal Analysis and Brand Review

Dell Technology is an American multinational technology company, one of the leading computers companies which is headquartered in Round Rock, Texas. In 1983, Michael Dell started selling computers which he later named Dell Computer Corporation. Since 1984, Dell has grown sprightly because of its distinctive direct-build-to-order sale model and operated through subsidiaries around the world. Since 1996, Dell products can be ordered online now. The key to Dell's success has been a fast and cost-effective supply chain which includes providing services to the customers by phone or website. In 2006, the company acquired manufacturer Alienware, and became partners with AMD (processors) and Intel. [Larsen., 2017](#); [Li., \(2022\)](#)

In September 2009, Dell took acquisition of IT service provider Perot Systems Corp for approximately \$3.9 billion. In March 2012, dell took acquisition of SonicWALL which specialized in firewalls and network security. It was formed in September 2016 along with the merger of Dell and EMC Corporation (later became Dell EMC). Dell was categorized 35th on the 2018 Fortune 500 list of America's largest companies by the entire business revenue. Dell also designs, sells and provides technical support for computers, servers, data storage devices, cloud computing, network switches, software, computer peripherals, HDTVs, cameras, printers and electronic equipment from other manufacturers. Dell has a worldwide reputation and fame for its products and superior customer service. The corporation has its focus on customer need and provides excellent support for any technical issues related to these products. Dell has expanded into storage

and network systems and is now looking to expand from offering pure computing to offering a variety of technologies to the customers which now focuses on launching mobile phones in the market. [Bierbrauer, A., \(2000\)](#)



Fig 1: Dell Logo and CEO- Michael Dell, Source: [Logmywaycorp.,\(2022\)](#)

1.1 Aims & Objectives of the Organization

Dell Technologies is committed to advancing human progress and being a leader in environmental, social and governance. The organization has a well-known reputation in the market, which is known for its products and services offered to customers. Dell's mission is to be the most fortunate computer company in the world at delivering the best customer support and experience. Dell is a competitive brand in the market as well as a brand name that people do not give second thoughts when it comes to purchasing the products. The company has a self-driven aim. The company is committed to driving human processes and leading in environmental, social and corporate governance. The company will meet customer expectations of the highest quality. Through products,

technology and people, the company seeks a positive and long-lasting influence by focusing on sustainability, cultivating and transforming lives around the world. In doing so, Dell will provide the highest quality of customer support and world-class leading technology. Instead of chasing dreams like usual crowd, the CEO wants the company to focus on areas of growth, including new projects like cloud computing, big data, security software and mobiles. Dell pursues a dominant strategy that offers the best value for money. The company grows its market share by charging lower prices while maintaining profitability through an effective direct selling model. In this research, I have talked about the organization's objectives, aims, strengths with a complete SWOT analysis. The research on this organization is based on secondary data collection and therefore, the approach, data collection and methods will be explained in the upcoming parts.

The above statement has the following below elements: Bierbrauer., (2000); Ragulan., (2021)

1.2 Mission (Objectives) & Vision Researchgatecorp., (2022)

A. Mission

1. Providing the finest customer experience: This is the mission that Dell put great stress and effort on providing the customers with products whose quality has been reviewed, tested and evinced. All Dell designs go through a rigorous process to ensure they meet the minimum excellence expected.

Ragulan., (2021)

2. Beyond assumptions: Dell is not satisfied with anything but providing its customers with the best experience. The company is committed to all the resources to make the best quality products and meet the changing trends of the market. Also, Dell makes sure to provide and adapt customizable systems to their customers. Dell will also enter innovation-focused partnerships to help develop IT skills through integrated educational programs. [Ragulan., \(2021\)](#)

B. Vision

Dell believes in constant change and how the business deals with the community. The focus remains on the customers, as the company believes that focusing on demanding customer needs is the changing future of technology. Dell's vision will be only towards customer satisfaction. As beneficial to become the successful leader in the computer market, Dell will fulfill all requirements to bring the latest technology and loyal customers. For that, the company does beyond everything to keep its customers happy. [Ragulan., \(2021\)](#)

The organization believes in four Moonshot goals: [DellTechnology.,\(2022\)](#)

- Promoting Sustainable Environment
- Cultivate inclusiveness
- Transform and reshape lives
- Uphold ethics & Privacy

In easy words, one can say that the aim of the company is to obviously make a lot of money and contribute with powerful, multi-featured computers which anyone can use without needing any technical support. Dell aims to provide all great computers with great

features including desktops, notebooks, tablets and mobiles that are accoutered with the prerequisite software. Michael Dell eventually became a partner with the supplier. This helped Michael cut out the mediator and order under the wire catalogues, direct sales and retail division. The core element of the plan of action revolves around five key direction keeps apart from the company's competition in the enterprise mobility space:

[DellTechnology., \(2022\)](#)

1. One-stop Solution
2. Global Support
3. Security
4. Manageability
5. Windows 10

However, the above points were kept in mind while launching Tablets but then after the results from the market, the company is going to keep the same strategy while launching mobiles. [DellTechnology., \(2022\)](#)

[Aims by 2050:](#) Although the first focus for the company is to make more money by launching new projects and getting more customers. But Dell also is an environmentally friendly company. As the author worked in the organization for 8 years, she found that the company is totally paper free, and everything is digital and on cloud. Also, by 2030, the company will reuse or recycle equivalents for every product which is purchased. Their packaging is made from 100% recycled or renewable materials. Below are some targeted points that the company will focus on. [Rissman, Bataille, Masanet,& Aden., \(2020\)](#)



1. To attain net zero greenhouse gas (GHG) emissions for extents 1, 2 and 3 by end of 2050. [Rissman, Bataille, Masanet,& Aden., \(2020\)](#)



2. Company focuses on the idea that by 2030, 75% of the electricity will come from geothermal sources at all Dell Technologies provision, and it will be 100% by the year 2040. [Rissman, Bataille, Masanet,& Aden., \(2020\)](#)



3. The brand will work with their direct raw material and distributors to decrease greenhouse gas emissions by 60% per unit of the total business profit by 2030. [Rissman, Bataille, Masanet,& Aden., \(2020\)](#)



4. Provide sustainable ability development for labors in the distribution chains every year along 2030. [Rissman, Bataille, Masanet,& Aden., \(2020\)](#)

1.3 SWOT Analysis (Strengths, Weakness, Opportunities & Threats/ Risks)

A. Strengths

1.Brand Image and Name- Dell has maintained a strong prestige as a global PC and IT company. Main reason behind this is the commitment to quality and innovation for the company's fame. Dell's reputation also can be imputed to its pricing strategy. It has added a substantial variety of commodities. [Sabbaghi, Vaidyanathan., \(2004\)](#)

2. Product Brochure- Dell has a large merchandise brochure that has been enlarged through acquisitions. The organization has divided the business into four categories including Infrastructure Solutions Group, Client Solutions Group, VMWare and Other Companies. The Client Solutions Group shared PCs, notebooks, branded peripherals such as monitors and projectors, as well as third-party software and peripherals. Dell has taken over EMC and merged it with Dell's Enterprise Solutions Group to form the Infrastructure Solutions Group. VMware provides enterprise computing, cloud, mobility, networking and security infrastructure software (Dell Annual Report., 2018). Other businesses incorporate RSA, Pivotal, Secure Works and Boomi. RSA and Secure Works offer security solutions. [Researchgatecorp.,\(2022\)](#)

3. Focus on Innovation- Dell is a pioneering and progressive brand, investing in innovation to continuously diversify the product lines. The company focuses on the development of adaptable technology solutions at fierce prices. The organization has started its interest in the investment in Research and Development. In 2016, 17 and 18, its investment was 1.1, 2.6 and 4.4 billion dollars. In 2018, it has firmly increased its investment in research and development. [Sabbaghi, Vaidyanathan., \(2004\)](#)

4.Global Augmentation- As mentioned above, Dell is a US-based company, but it has now expanded its business in other countries including Europe, the Middle East, Asia and other geological regions. Moving forward with Japan, Canada and Europe. China and India are among the mushrooming markets currently. [Researchgatecorp.,\(2022\)](#)

B. Weakness: This can include several factors, here company must focus on what can be improved while moving on to this project. Any internal deficiency that can hinder the company in meeting those stipulations. The organization must check if they have the proper instruments and tools as they are focusing on launching a completely new project. [Sabbaghi, Vaidyanathan., \(2004\)](#)

C. Threats/Risks: Any risk in the project is an unassertive state, that can have any positive or negative impression on the performance or the overall result of the project. Here applying risk management techniques can possibly provide solutions to the upcoming problem. It includes planning for the risk, identifying and analyzing them. [Hillson, D. and Simon, P., \(1999\)](#)

1.High Business expenses-Where the company is leading in the world, there are some business expenses that have caused severe losses as well. The organization's operating expenses increased to \$23.4 billion in 2018, resulting in a dissipation of \$3.3 billion.

[Hoffmann, Petit, Bostoen, Heiden., \(2022\)](#)

2. Surviving Market change- This is one of the biggest risks that this company must face among other brands in the market. There are already brands who make remarkable mobiles, and they already have loyal customers who probably would not like to change, especially apple users don't like to go for android platforms. Dell's computers are

undoubtedly remarkable, but will the mobile phone make the same impact is significantly a risk for the company.

Hoffmann, Petit, Bostoen, Heiden., (2022)

3. Completely dependent on the US market- The company earns a huge part from the US market and nearly 48% of income comes from the US market. It has been trying to increase the presence in Asia/Pacific but still is strongly dependent on the US market.

Hoffmann, Petit, Bostoen, Heiden., (2022)

4. Looming Recession- As the world moderately returns to its economies, profitable wreckage and loss of jobs have pushed several countries towards descending volution and into recession. Hoffmann, Petit, Bostoen, Heiden., (2022)

There were several other threats related to this project that can bring the company towards downfall and the organization must be prepared for the upcoming risks and challenges. Here are some risks categorized based on this project:

Organizational Risks	Occupational Risks
Brand focused crowd: Apple and other competitors have already a name in the mobile industry and people prefer these mobiles phones. Although Dell is a globally known brand in the mobile phone industry there are other market leaders preferred for their phones. Author., Gagandeep Kaur (2023)	Health risks: As per the recent research. People who work in long shift tend to have risks of more health concerns as compared to other jobs. It could be mental stress, high blood pressure, severe headache. Author., Gagandeep Kaur (2023)
Cost risk: Money invested in this project will always be a great risk. One never knows if this project can be a success or failure but can keep working on loopholes and trying their best. Author., Gagandeep Kaur (2023)	Mental Wellness: Staying in offices while working on major projects and working all day will lead to no social life and will lead to poor mental health. This can also cause depression and severe lonely feelings. Author., Gagandeep Kaur (2023)
Skills resource risk: For this project, the company needed to hire skilled and competent employees, those who already have a good knowledge of this field and experience otherwise it will cause severe damage to the project. Author., Gagandeep Kaur (2023)	Willingness to work: One of the major risks to not being willing to come to work. This can be possible due to the poor work environment. People have the tendency to mock or bully others at the workplace, which can affect one's health and can

	affect overall project performance. Author., Gagandeep Kaur (2023)
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Table1: Risks in the project., Source: Author., Gagandeep Kaur (2023)

The above risks can be avoided by focusing on one-one meetings with all employees and making sure, company is meeting the required target. The organization must make sure to focus on the targeted project but also needs to keep an eye to check what's going on in the market. Making sure that every employee is being provided with a mental health consultation. Author., Gagandeep Kaur., (2023)

C. Opportunities

1. Focusing on surfacing economies- Opportunities in emerging markets are expanding rapidly as evolved markets reach maturity and saturation. First, Dell can offer commercial explanations in countries like Africa to take benefit of the expanding demand for cloud computing services along with this new project of mobile phones.

Sabbaghi, Vaidyanathan., (2004)

2. Increase in Acquisitions- Most of the significant companies are enlarging themselves swiftly into various markets and regions through acquisitions. Dell has the resources to fund extensive acquisitions to rapidly scale innovative start-ups and companies.

Sabbaghi, Vaidyanathan., (2004)

3. Expand enlarging segments- Demand for smartphones endures elevated and is expected to grow in the forthcoming years as mobile usage increases. Dell is working on this segment to expand the business and to exploit this demand.

Sabbaghi, Vaidyanathan., (2004)

1.4 Dell's Strategy of Recruiting Employees and Developing leadership qualities

Dell knows exactly where to position the right staff with the right set of skills & training. The company has publicly announced that by 2030, 50% of its global workforce and 40% of the organization's staff will be women. Company strongly believes on the below targeted points explained in brief: Cone., 2000; DellTechnology., (2022)

1. Let the Candidates Get to Know You- Dell's technology tactic is to give candidates a glimpse of the company. It helps candidates validate themselves and those who fit can invest more in the company. The company's careers site has several opportunities to introduce candidates to the work culture at Dell. As soon one goes to the webpage, four items are as listed:

- An illustration of the Dell mindset
- Buttons for searching and applying for available jobs for all types of candidates
- As the website is categorized into sections, The next section when you scroll down covers nine topics including:
 - The People
 - The Workplace
 - The Company History
 - Culture Code
 - Employee Benefits and Rewards & developing Programs; Cone., (2000)

2. Candidates and Clients are One Group- Organization acknowledges that two efforts can be rolled into one: customer acquisition and employee acquisition. Current or potential future customers can influence their point of view towards the company if they are referred for a job or an interview with the company. Dell knows how to find out the leader from the crowd, either it is the one being in hiring process or the one from within the organization. Employees working inside the organization are way confident and it shows in their physical gestures and way of talking.

Cone., (2000); DellTechnology., (2022)

3. Defining the roles and responsibilities- Dell defines the company culture by their values and is brought to life by the way the employees work and lead every day. The organization always prefers to promote insider employees with extraordinary performance and confidence. All the senior leaders have the qualities of being a role model and a friend to the team members. There is always competition between old employees and new employees at the time of appraisals depending upon their qualities as the organization always interviews and hire people against their competencies. This process cuts out costs to the company and speeds up the process of recruitment and allows time to save some time and money for the company.

Cone., (2022); DellTechnology., (2022); Recruteze., (2022)

CHAPTER 2. Background Research

Discussing the products that Round Rock-based tech giant Dell Technologies has been working on, he aims at everything from the future of hybrid work. The company is so customer focused that it keeps on staying up to date as per customer needs. It also uses different ways to find out customer satisfaction, like reviews and feedback via emails, phone calls. Dell also relies on Net promoter scores to find out if customers would like to refer this brand to their friends and family. The company follows Agile and 6 sigma methodology to run the projects as it keeps it running on track on time and it involves constant collaboration with the stakeholders and improvement on every stage which is what this organization focuses on to make sure the customers are their priority and their satisfaction is what is focused on. Also, Dell works using six sigma as it uses statistics and data analysis to analyze and reduce the defects in all the projects this company holds on. Talking about the background and analyses, below characteristics of the organization have been researched in brief

Author., Gagandeep Kaur (2023); DellTechnology., (2022)

2.1 Industry Trends- It has furthermore sub parts in it: -

- A. Digital Reshaping: Digital reshaping or transformation is the process of using digital technologies to create new business cases, cultures and customer experiences, or to alter the old ones and meet the new changes of business and market. In recent years, the company has begun to see this economic environment from a macro perspective, and it has been beginning to shift to an environment

that emphasizes cost reduction, and which leads to digital acceleration.

Holmström.,(2022)

B. Automation/ Artificial Intelligence or use of Machine Learning in Industry:

Automation is no longer confined to just one industry. Automated feeds can now be seen in every industry, from manufacturing to retail to healthcare to agricultural services. Artificial intelligence (AI) technology is being used in an increasingly varying set of organizational implementations, creating new types of people as machine configuration plays an increasingly important role in many aspects of modern organization such as administrative directing, design, and manufacture.

Holmström.,(2022)

C. Metaverse: XR (augmented reality), AR (artificial reality) and VR (virtual reality)

technologies are now known to everyone and have been amidst and are known from time to time. But also, there is also deep interest seen between end user and customer in adapting these Technologies, and the actual experience is getting better. Holmström., (2022)

D. Sustainability: Many customers and end users are now interested in how Dell can do more for sustainability, the carbon footprint and the desired wider impact.

Holmström., (2022)

2.2 End-user-trends

According to Matt Baker, senior vice president of corporate strategy at Dell Technologies, the end of 2022 marks a turning point for the next wave of technology.

“There is no singular workplace. Work is now an outcome, not a single location.” –Matt Baker, VP of corporate strategy, Dell Technologies

In this recent interview, chief information officer Jen Felch talked about the blockages to the digital transformation and multiple opportunities and below are some of the takeaways mentioned:

Hybrid by design: As company mentioned that due to pandemic the work was brought remotely as the situation required it, but now it has created an opportunity to keep it hybrid by design. As the company believes that they are no longer a single location but now an outcome, they have now adopted this hybrid model. And combining this hybrid model with Artificial Intelligence, they have now employee morale and more productivity.

[DellTechnology.](#), (2022)

Cybersecurity and Triple A method: As mentioned earlier, remote work has been preferred by many global employees without any challenge. But this also means that the chances of cyberattacks, and malware are more vulnerable now. And to overcome this challenge, organizational leaders considered the Triple A method, which includes Authorization, Authentication and Accounting. [DellTechnology.](#), (2022)

Broadening the ecosystem: Dell experts believe that a gaping ecosystem will replace the direct pattern of the past and will harbor a new era of technological advancement and digital transformation. The importance of this open ecosystem is critical to the company's operational diligence today. The organization cannot maintain the same availability and resilience as when everything was in the house. [DellTechnology.](#), (2022)

2.3 Similar transformations in other businesses or organizations or sectors

All these multinational organizations work toward the evolution of Technology and (Information Technology) is made up of many interconnected physical systems, splited by a set of cables. Focus area was always the space in which these systems are designed. While going through digital transformation, now IT consists of multi-cloud environments and the applications and associated data that is constructed. While working on mobile technologies, companies focus on the various areas to attract more customers. Mobile marketing is an interconnect digital advertising strategy which is focused at reaching audiences via smartphones, tablets and other mobile devices, websites, email, SMS and MMS, social media and apps. In recent years, customers have all their focus and concentration on mobile phones. [DellTechnology., \(2022\)](#)

IT is changing so quickly that departments hustle with the budgets while supporting all the efforts. Surveys have found that the road to digital transformation presents many challenges and obstacles that can divert valuable IT capitalize from more innovative initiatives. As an example, now-a-days malware and ransomware viruses have invested in robust cybersecurity solutions. Problems have given birth to solutions themselves here. All this discussion is just about customer needs, Innovation, trust and scalability to achieve IT optimization goals. Whereas MNC's like Dell are working hard to get whatever keeps their customers happy and satisfied, customers also must trust their suppliers and partners by keeping an eye on the innovation which is just around the corner and always ready to scale. To make sure, whether it is mobile industry of computer market, the chart should rise every quarter and brand name should be in the top of the market. [DellTechnology., \(2022\)](#)

2.4 Key trends in mobile communications in 2023

1. Global Adoption of 5G- The Survey Report for June 2022 shows that around 580 million mobile customers will opt for a 5G subscription by the end of 2021. That number is expected to reach 1 billion in 2022 and 4.4 billion in 2027. The global rollout of 5G offers many advantages over 4G. It offers faster connections, sub-millisecond latency, and superior user experience and functionality. [DellTechnology., \(2022\)](#)

2. Cumulative Web Apps- Progressive Web Apps (PWA), also known as hybrid apps, are used to develop web technologies that will become the trend for mobile apps in the future. They are inexpensive and cost effective to build and operate offline. It is also linkable (and allocatable). These apps can be optimized for search engines and users don't even need to discover app marketing. These apps need less storage on mobile phones compared to pre-installed apps. [DellTechnology., \(2022\)](#)

3. Mobile payments- Payment apps have been throughout for a long time and the market is now strong and diversified and assorted with considerable solutions from banks, operators and independent service providers contributing to steady growth. PayPal, Google Pay, Apple pay, Venmo, Cash App are a few examples. These apps are more convenient and faster to operate and save one's time and one can operate them while just sitting at home. Mobile payments eliminate the need to carry credit cards and cash. In case, If the smartphone is stolen, sensitive data is protected as per the data protection policy of these apps. [DellTechnology., \(2022\)](#)

4. M-Commerce- Mobile commerce is the method of purchasing and vending goods and services using wireless portable devices such as smartphones and tablets. M-commerce

is a form of e-commerce that allows users to acquire online shopping platforms without using a computer. Due to pandemic and social distancing scenario, mobile commerce sales have revolted in the market. According to surveys, this value might reach \$600 billion by 2024. [DellTechnology., \(2022\)](#)

5. Foldable Electronic gadgets- If digitalization is categorized, foldable devices will remarkably become a part of it and will increase between the years 2023-2027. The size of the market is expected to grow by more than \$70 billion. So far foldable devices have been the strongest and best trend in the mobile estate market. It has become so appealing that the largest screen will still fit the pocket of a user. These gadgets are a boon for the growth of the business and since Dell thought of launching these devices in the market, business will rapidly grow as per the assumptions. [DellTechnology., \(2022\)](#)

6. Mobile Augmented Reality- Mobile Augmented Reality is an AR service developed for mobile devices. People can communicate with simulated objects on mobile screens that mix real-world environments with 3D virtual objects. This is the technology that has already brought some great features and truly immersive experiences to e-commerce mobile apps. Like chasing Pokémon and real monsters with the Witcher. It will be way different by 2023 and beyond with the dedicated Augmented reality Hardware. [DellTechnology., \(2022\)](#)

9. Mobile Security: Two-step Authentication, Biometrics- With core apps containing multiple modules (mini apps), a single breach can result in horrific data loss, identity theft, credit card number theft, unauthorized access to other services, and more. Now-a-days apps rely more on security measures reserved for professional services, like ID

verification, two step authentication and even includes personal biometrics like facial recognition or fingerprint identification. [DellTechnology., \(2022\)](#)

10. Artificial Intelligence & machine Learning- Artificial intelligence and machine learning are essential to optimizing the user experience and ultimately delivering the best customization of mobile apps. Chatbots are becoming increasingly popular in mobile apps that provide customer support. According to research, their market alone (not just mobile apps) is expected to reach \$13.9 billion by 2027. [DellTechnology., \(2022\)](#)

2.5 Data Analytics in the Organization

Data analytics is the application of technology to obtain useful and beneficial information from data, including big data. Data analysis techniques and processes have been automated into mechanical processes and innovations that influence crude data for human consumption. Why is Dell using Data Analytics in this project, answer is simple, as it clearly gives us the results. It is explained as below by its four types of [Sedkaoui, S., \(2018\)](#)

1. Descriptive Analytics: Explains us what happened
2. Diagnostic Analytics- Explains why it happened
3. Predictive Analytics- Explains what likely will happen in the future
4. Prescriptive Analytics- Explains us how to act

[Sedkaoui, S., \(2018\)](#)

Increase in data generated by all multinationals, individuals, scientists and all public officials in Cooperation with further development of IT tools provides a new systematic

viewpoint. This will help Dell in managing customer data while going through this new project. There are some advantages for the organization:

1. Keeping up the decisions- Any organization can take advantage of the large amount of data they need for their projects. One must filter the data according to the requirements and see which data fits here. [Sedkaoui, S., \(2018\)](#)
2. Price pruning- Big data has been proven to be very useful in extension of the company's existing architecture. Therefore, with the conventional use of analytics, projects can be highly successful in cutting operating costs and prices. This is usually one of the biggest challenges. [Sedkaoui, S., \(2018\)](#)

2.6 Comparing different perspective in the Background Research

Dell believes it is all about emerging change and technology is emerging along with this which feels strange but filled with curiosity. Mobiles phones have taken over the life of a common man. Mobiles phones now can do anything that a human can think of. Below are some different aspects that this organization will come with that will have a different impact on the economic development in some rural countries like Africa and expecting some development changes. [DellTechnology., \(2022\)](#); [Brian Mullins., \(2017\)](#).

1. Digital Conductors: Technology has been so advanced that now-a days all the mobility needs are fulfilled by just one gadget. Sooner it will be very difficult to justify keeping a piece of instrument that is hardly used. Everything now-a-days. Is being monitored by an app or a gadget. Mobile phones are way easier to carry with all this information. They are not only connecting people in need of driving, or job search, for vacations, or with any critical services. By 2030, Technology will add-

on as humans and helps Coordinate, manage and automate many routine tasks activity. By 2030, many skilled digital orchestra conductors will rely on a range of personal technologies, including voice-enabled connected devices, wearables, and implantable devices to extract and activate intentions from patterns and relationships and deactivate the resource accordingly. [Brian Mullins., \(2017\).](#)

2. Work chasing people: Human-gadget partnerships not only help with automation, but also adjust our lives and change the way this organization works. With helping in finding talent, managing teams, delivering products and upshots and services and support professional evolution. Work chasing people will also help in minimizing personal biases and pigeonholes in job seeking processes without revealing their gender or ethnicity. [Brian Mullins., \(2017\).](#)
3. Ethnic & Cultural Empowerment: Dell Technologies recognizes that companies committed to continuous growth and innovation must be equally committed to equality and inclusion. This is one of the many reasons that the organization is working to build a workforce that reflects the diversity of the customers, stakeholders and partners around the globe. [Mullins.L.,\(2017\); DellTechnologies., \(2022\)](#)
4. Mobile Phones & Economic Developemnt in Africa: Mobile phone access and use in sub-Saharan Africa has increased dramatically over the past decade. Mobile phones have brought new opportunities to the continent. Mobile phones connect individuals, information, markets and services, cutting the gap between urban and rural areas and between rich and poor. Mobile phones have greatly reduced communication costs, allowing individuals and businesses to send and receive

information on a wide variety of economic, social and political issues quickly and inexpensively. New research shows that reducing communication costs associated with mobile phones can lead to tangible economic gains by improving agricultural and labor market regulation and improving producer and consumer well-being in certain situations and countries like Africa. [Aker and Mbiti., \(2010\)](#)

2.7 Evaluating the relevance, reliability and validity of the different sources of background research

Let's consider all these points one by one:

1. Relevance of background research: This research has given me more brainstorming, so I had to pick this topic. As an author while writing this thesis, I realized that working with an organization like Dell Technologies has given me immense experience and exposure that I can write about the company about the experience of working with them. I worked with this organization in the consumer segment and was already aware how the company is focused on customer driven goals and projects. When I started my secondary search, I came up with so many articles and journals and blogs that made me realize that the organization is vaster than my thoughts on this research and I could explore every section I am supposed to and write briefly on it. Dell is one of the leading companies and has many current and many upcoming projects on the list and one of them is getting back into the business of mobile phones. This research has every detailed point covered as needed. It includes a brief about the company, the background, the strategies, future and upcoming projects. [Author., Gagandeep Kaur \(2023\)](#)

2. Reliability of background research: This research is a self-completion questionnaire with secondary data collection and perhaps the most preferred way of collecting quantitative data. This study intends to focus on the new project of Dell where the company is focusing on launching mobile phones on the market with the possibility of an increase in revenue. Data is collected from all the reliable sources, articles, journals, websites and blogs and has been truest to the best of my knowledge. And yes, the study does include all the information about this organization working towards the new project. Here the root of the research is the new project that this organization is planning to launch, and we have discussed about its scope, and the extent to which the studies has successfully investigated that how effective it will be for the organization and there are chances of an increase in the revenue and brand name for the company. [Author., Gagandeep Kaur \(2023\)](#)
3. Validity of background research: In this research I have focused on 4 points to make sure my research is valid.
- 1) Internal Validity: Internal validity is defined as the extent to which the observed results for this research have represented the truth of the data under study and is therefore not due to methodological errors.
 - 2) External Validity: The purpose of scientific research is to produce generalizable knowledge about the real world. It also includes how this research was relevant and applied to a broader context.
 - 3) Construct Validity: It has successfully explained how this research was going and what is claimed in this research as what was intended to be claimed.

- 4) Statistical validity: Here I have used all the reliable data and to the extent to which conclusions drawn from research studies obtained from statistical tests are considered accurate and reliable. Author., Gagandeep Kaur (2023)

2.8 Identifying connections between the background and current knowledge of the subject using different strategies

Here the background is clearly linked to the current knowledge as they both rely on each other. We can co-relate the opportunities, strengths and threats.

Opportunities	Strengths-Opportunities	Threats-Opportunities
1.Developing economy	1. Plan of action for the developing economies	1.Increase R&D on PC's and other products
2.Growing online trading	2.Using customization combine in all the possible products	2.Continual reduction of costs and increased sales by focusing on an online sale plan.
3. Immerging with new technologies such as cloud storage and mobile phones.	3.Maintaining brand name & keeping same strategies for maintaining old customers	3.Agreeing to unnecessary customer demands

Table 2: Corelating opportunities, strengths and threats; Author., Gagandeep Kaur (2023)

2.9 Relevance of Background research with the Business Project

In today's competitive era where global marketing is growing every day, consumer perceptions of the country of origin of a product are shifting from one new perspective to another. The growth of electronic devices such as mobile phones globally is the most relevant for this research. But here the discussion will focus on Dell making mobile phones and selling them all over the world. Talking about the current mobile phone market, it was studied that there are 32 different attributes that consumers value when purchasing a mobile phone. From this analysis, nine key decision benchmarks are derived. These criteria are used to classify this consumer market into four behavioral collections: practical, rigorous, value-conscious and fascinating. Young adults or the younger generation is crazy for the mobile phones. It could be Apple or Android platform, they love all. Now-a-days mobile phone market is one of the leading markets in the world. Everything is now compacted in our pockets and mobiles have everything that an individual needs. This is the best reason for this organization to launch a new project where the purpose is served for all and the organization gains name recognition and revenue growth. [Author., Gagandeep Kaur \(2023\)](#)

Findings: Practical consumers are the ones who give high importance to the functionality, physical and convenience-based attributes of the mobile. This group also prioritizes functionality along with design. While value-oriented consumers are very price conscious, the mesmerizing segment is a group of all-rounders who value many attributes such as technical superiority, practicality, durability, functionality and design. [Author., Gagandeep Kaur \(2023\)](#)

Research constraints/ Implications: High-tech products such as mobile phones are becoming an integral part of people's lives. It leads to a more involved decision-making

process. It is important for marketers of these products to understand and address the behaviorally diverse consumer segments that exhibit significant differences in their decision criteria for such products. Author., Gagandeep Kaur (2023)

CHAPTER 3. Method

As discussed above, the main motive of the company is to launch new products in the market and increase the chances of brand name and increased revenue. Also, the organization is environmentally friendly and continuously focuses on recycling and reusing. In upcoming years, the company prefers to go 100% recyclable and reduce greenhouse gas emissions. The organization continuously works on customer needs and making sure this should not impact the environment. Dell as an organization works on Agile and 6 sigma methodology as it is easy for the organization to keep all projects on track. Dell works on every individual and interacts and review every project and process with tools; working software over rigid and comprehensive documentation; customer collaboration over the negotiation and then responding to a following plan, which makes it easier for the company to work and keep it on track.

In this research, the author has used and explained the QDAS method to analyze the qualitative data for the research. In the last 25 years since the Qualitative Data Analysis (QDAS) software program was developed, the use of such tools has been confirmed as both a positive and concerning development for the field of qualitative research. When the primitive QDAS applications were developed in the 1980s, researchers hypothesized that such tools could feasibly advance qualitative research in several keyways. There were many ways to analyze the data, the first was to extend the paper-based method of

data encoding, retrieval and analysis of what was beyond the limit and feasible by hand by enabling more complex, and adaptable coding schemes for more accurate data retrieval. A second possibility was that QDAS could improve qualitative analysis and clarification by supporting a form of analysis that could not be performed manually by linking a text or sound, graphics and video. [Richards & Richards., \(1991\)](#)

The author conducted a qualitative content analysis of several pragmatic studies following articles and journals between 2018 and 2022. Then, restricted the search to proper program names (ATLAS.ti™, NVivo™, NUD*IST™). I, then tapered down these search sets to re-evaluated journals and articles published in English as this was the prepotent language used for research. The final focus was only on practical rather than methodological articles, especially since the focus was on how researchers combine QDAS into their research practice and reporting. There were program codes used to arrange the types of research study administered and types of data accumulated and assembled and stages of the research process when this QDAS was used (literature review, data collection/ analysis and management and findings). This included two stages. [Richards & Richards., \(1991\)](#)

Stage1 included developing and handling the strategies with the help of Excel spreadsheets to arrange all articles in data set and one more reason is that excel is easy for everyone to read. Each data set was then separated and categorized into easy sub parts to check and make sure, we are still reading data in qualitative form, and which is easy to read like an article and not in any coded form. But at the end of the stage, there were findings with the limitations of use of excel as working and going through this research and going through 300+ articles, spreadsheets were unwieldy, when tried to

review the entire dataset. By stage 2, the previous analyses articles were closely monitored and undertaken using NVivo (software program used for qualitative and mixed method research). Below is the type of research studies conducted with QDAS support.

Richards & Richards., (1991)

ATLAS NVivo Total						
Research Type	Articles	%	Articles	%	Articles	%
Generic Qualitative	28	8.1	150	36.7	178	23.7
Focus groups	73	21.22	40.1	9.71	114	14.7
Grounded Theory	45	12.9	55	13.3	100	13.11
Content Analysis	15	4.7	33	7.8	49	6.5
Case study	20	4.9	21	5.7	42	5.5
Survey Research	14	3.9	8	1.9	20	2.7

Table 3: Research Studies concluded with QDAS support; Richards & Richards., (1991)

ATLAS NVivo Total						
Data Type	Articles	%	Articles	%	Articles	%
Focus Groups	111	33.3	64	15.6	179	23.8
Documents	38	9.6	55	13.5	93	23.7
Surveys or questionnaires	28	7.5	52	11.9	78	10.2
Image data	5	1.5	10	2.7	27	3.3
Online media data	6	1.5	7	1.8	13	1.7
Other	7	1.7	4	1.2	10	1.6
Websites	1	1.1	7	1.4	7	0.9

Table 4: Data Types Used; Richards & Richards., (1991)

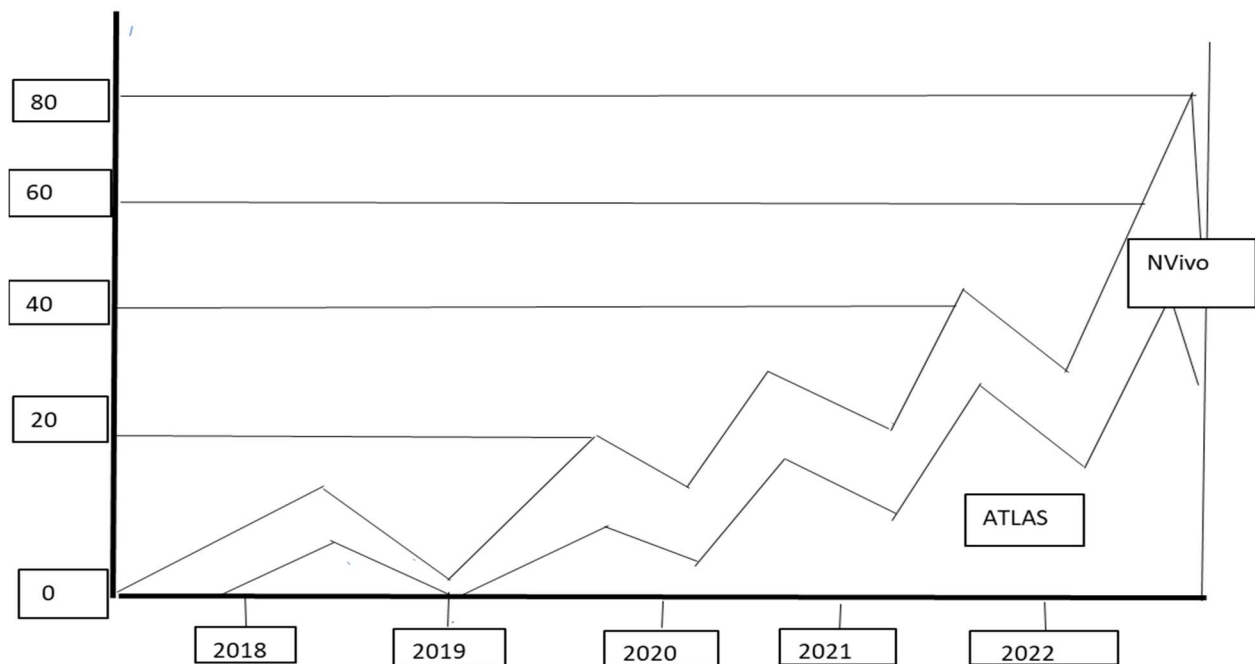


Chart 1: Quantitative data analysis between year 2018-2022., Author., Gagandeep Kaur (2023); Richards & Richards., (1991)

Findings: From this section of report on the users and use of QDAS as detailed in multiple articles published between 2018-2022 using ATLAS and NVivo. In fig 1, it shows how the articles using QDSA have increased each year over the previous one. From 2018, there is a 50% increase in the number of articles published on related topics and data was easily determined. Richards & Richards., (1991).

3.1 GAP in the research

This research is a part of Dell Technology's upcoming project to launch mobile phones. If the project is led right, it helps every part of the business run more smoothly. It authorizes the team to concentrate more on the work which is important. It helps and sets it free from the distractions and intrusions caused by tasks getting off track or budgets going out of control. Everything the company is working on is farther than expected and extraordinary

in terms of acquiring the computer market and going towards the mobile industry, this organization is on fire. But there are a few gaps that I could not fill up with. [Author., Gagandeep Kaur \(2023\)](#)

1. No Projects, No Jobs: In easy words, if the company, Dell Technologies, has no new upcoming projects, employees had better be looking for new jobs. The organization is doing good so far with the computer and peripherals business, but launching new projects provides job opportunities to new people and inside company appraisals as well, that gets life going. [Author., Gagandeep Kaur \(2023\)](#)
2. Zero Quality Improvement: This also gives the organization a chance to continuously work on improving the product quality. This will surely increase the customer satisfaction rate and will bring glory to the brand name. But, if there are no new projects, the company will not know what to do in case of improvement. Launching this new project is giving the organization a chance to study the learning from this. [Author., Gagandeep Kaur \(2023\)](#)
3. Poor error reduction rate: This project was launched earlier as well but the company was more focused on the computer market then, so, it was not a success. But, as the company is prepared for it, this will reduce the error rate. If this project is not launched, the organization will not know how to improve the weak points which can bring profit to the organization. Continuous improvement will bring more success and more success leads to more projects which eventually leads to hiring and giving jobs to more people. [Author., Gagandeep Kaur \(2023\)](#)
4. No New customers: Launching mobile phones on the market will bring a greater number of customers. Dell is the brand name and people buy it for the best

customer support. If this project is not launched, consumers will not know that this company can also make good mobile phones and will miss out on good products.

Author., Gagandeep Kaur (2023)

5. Zero opportunity to grow in foreign land: This is the first time that the organization plans to launch mobile sin Germany & other European countries. If the project is not launched because of the fear of previous failure, one might never know the opportunities this company will miss out. Germany is one of the successful countries and bringing a new project in this place will immensely increase the profit.

Author., Gagandeep Kaur (2023)

3.2 Research Approach & Rationale

As per PRINCE2 (2009), every research approach is the process of explaining the work and how it would be performed and actioned. Moreover, regulating a research study requires exceptional expertise to make it meaningful and impactful. This approach includes the stages below of the project. Carberry., (2021); Hendriarto., (2021)

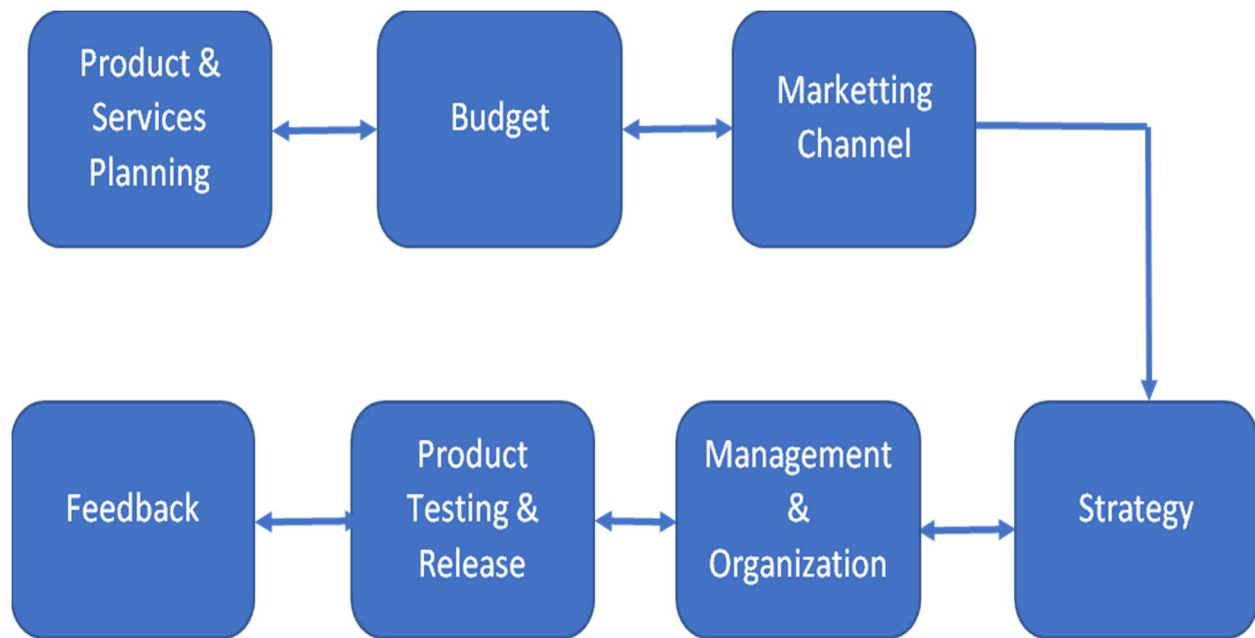


Chart 2: Agile and Prince Approach on the project; Barrett, D. and Twycross, A. (2018)

1. Product & Services Planning: This part includes the products and the services which will be provided to the customer, which includes Dell Mobile phones with a focus on how the consumer will like it and will benefit from it. PRINCE2., (2009)
2. Budget: Launching a new project will include a lot of investment. This organization is already a brand name and has developed its roots in the electronics market, so there should not be any problem in investing in new projects but keeping finances on track. PRINCE2., (2009)
3. Marketing channel: This step concludes the market analysis, which will contain the details of a fair idea of the targeted consumer segment, with an overview of the industry, the historical and current market forecast and competitors. It should be backed up by quantitative and qualitative data collection. Social media, like Facebook, Instagram, Emails and YouTube can play a vital role in marketing new product advertisements. PRINCE2., (2009)

4. Strategy: By combining all the information, demonstrate a strategic meaning of reach of the desired goal from the promotion of the product to the cost of the labor to other operational part including management and lifecycle. One new strategy of the organization was to keep the price of mobiles less than the other companies, so that it would be economical for everyone. PRINCE2., (2009)
5. Management & Organization: Explaining the roles and responsibilities of the employees and team members. All the consultants like accountants and attorneys of the organization will be listed here. Stakeholders will be included too with a description of the percentage of their ownership. PRINCE2., (2009)
6. Product testing & Release: Final Stage is testing the mobile before its launch. Checking with the testing team and making sure it is ready and fit for the launch. After this a final date is chosen to release the product on the market.
7. Feedback: Taking feedback from the customers, which will include keeping track of if any of the consumers complained about any products sold to them. Keeping them in loop via emails and asking to share continuous feedback to do better. DellTechnology., (2022); Barrett, D. and Twycross, A. (2018)

3.3 Research Instrument and Data collection

Data collection is the technique of collecting information about a specific product. It is very crucial to ensure that the data is completed during the collection or draft phase, and it is collected legally and ethically. In general, there are three types of consumer data that Dell as an organization collects via different ways.

Author., Gagandeep Kaur (2023)

1. First party data: which is collected directly from the customers by Dell.
2. Second-party data: which is shared by another organization about its customer (or its first party data)
3. Third-party data: This data has been assembled and sold by the organizations that don't have a connection to the organization or the customers.

This data for the research has been collectively gathered from the different blogs of the websites, articles and journals. Dell has now taken over EMC, Alienware, SecureWorks and Virtustream. Dell operates in more than 180 countries now and now one can imagine the number of customers this organization has. In the fourth quarter of 2022, Dell held a 16 percent share of the global PC unit shipment in the market. As Dell itself collects its customers' feedback and suggestions via emails, phone call surveys, in a similar way, this data was collected from Dell. The country has different blogs where every update for the organization is available. It was very helpful in this research as all information was easily accessible. This is one of the biggest advantages of secondary data collection.

Author., Gagandeep Kaur (2023)

3.4 Secondary Data Analysis with example

In secondary data analysis, here author has used information collected by someone else for their own purposes. By using secondary data analysis, it provides a chance to answer new research questions or examine alternative perspectives on the original question of previous research. Devine., (2003)

Sample: Let's consider an example, if a Project Manager in any new business case is considering opening a new business, he could use the poll data that has been

accumulated by the government. Although the Project Manager would not collect the data himself, as the poll data includes information that could greatly benefit him. It could include average age, household income and literacy level in a particular region. By digging into this poll data to inform the decision of whether the Project Manager should open the new business, he has performed the secondary data analysis.

Devine., (2003)

3.5 Factors that were considered before conducting Secondary data Analysis

There are certain factors that were kept in mind before deciding whether to proceed with secondary data analysis. Author., Gagandeep Kaur (2023)

1. Learning about how the data will be collected.
2. Learning about the population that is already using Dell products
3. Learning what the objective or the original study was
4. Determine the response category for each question presented to survey respondents.
5. Determines whether clusters or stratification should be considered when analyzing data.

3.6 Advantages of the Secondary data Analysis

1. One of the most distinguishable advantages of using the secondary data analysis is its cost effectiveness. Researchers don't have to spend money, time, or effort on the data-gathering phase of a study because someone else has already collected the data and all

it is required is to collect the data properly from the sources. Secondary data may have to be purchased by the researcher if they wish to use it to inform the research they are working on, but this cost is almost always covered by the same data set, less than it would cost to create from scratch. [Author., Gagandeep Kaur \(2023\)](#)

2. And because data from secondary datasets are typically already scrubbed and stored in electronic form, researchers can keep their minds on and analyze the data rather than spend time preparing it for analysis. Also, collecting Primary data is little difficult as sometimes when one is on research about a product, not every company is open to share the details with him, so collecting them from the website and blogs is easily available and true to knowledge. [Author., Gagandeep Kaur \(2023\)](#)

3.7 Ethics

Research ethics are important for several reasons. They facilitate research goals such as expanding knowledge. They uphold the values necessary for cooperation, such as mutual respect and fairness.

Arden University has very crucial ethics when it comes to studies. As per the University, ethics are a vital part of the study system. Here, everything is an open book, but one must find the correct way to use the information in their own words and knowledge. All this data is stored in one drive and cloud by the university and is only accessible to the people who have authorized permission. The author also is a student of this university and has the authorization to store the data here.

[Stankovic. M, & Stojanovic, 3rd ed, \(2020\)](#)

There are several reasons why it is important to adhere to ethical standards in research. First, standards promote research goals such as knowledge, truth, and error prevention. Research ethics are crucially important when it comes to the research project. On the one hand, it helps facilitate general research goals, such as the quest for knowledge and the intention to avoid mistakes. Second, ethics promotes the beliefs for successful alliance, such as respect, trust, and accountability. Here for this project idea, the core idea was first discussed with the supervisor and when it was assured that it would be a good topic to research about. At the end of this thesis, the approval document has also been attached. [Stankovic. M, & Stojanovic, 3rd ed, \(2020\)](#)

CHAPTER 4. Results & Findings

This research is all about Dell Technologies and its upcoming project of launching mobile phones in the market which could possibly lead to an increase in the revenue of the organization. Mobile phones and smartphones are becoming more and more of a center of attraction and communication devices in everyone's lives. We believe that mobile phones will transform many areas of the financial system including business, healthcare, social networking, environmental monitoring, education and transportation. As per the news article, Dell is sooner going to enter the smartphone market. According to a journal Dell has been the best in the computer industry so far but the competition will increase with the launch of mobile phones as there are already competitors who are already experts in this field. According to the report, Dell had engineers working on smartphone prototypes for a long time now in Chicago. This site worked on Google's Android platform as well as Windows platform mobiles. But the organization is still not committing to any news yet, as it might not announce any news until it is confirmed and launched already.

By 2012, the organization stopped selling smartphones in the U.S as the company was then more focused on emerging markets and higher margin products. Earlier the company launched Venue and Venue pro smartphones and after that no replacements were announced. But they continued to be sold overseas. From the below chart we can see the top mobile phone companies in the market so far: [IDCcorporate., \(2022\)](#)

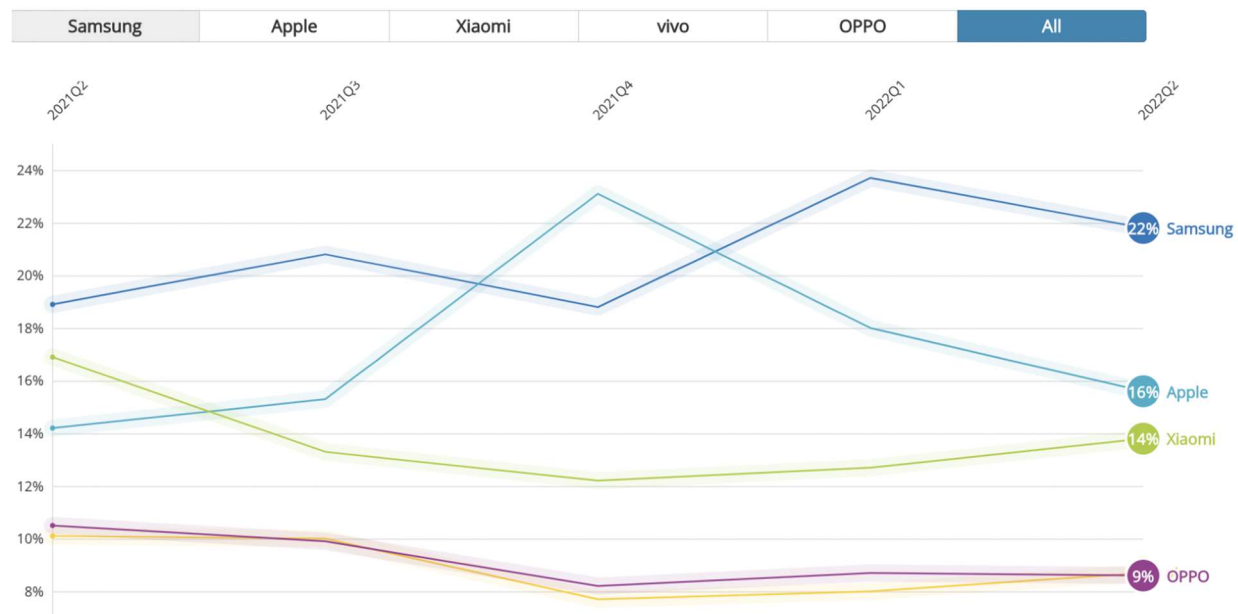


Fig 1: Smartphone market share, Source: [IDC corporate., \(2022\)](#)

Company	2022Q2 Shipment vol.	2022Q2 Market share	2021Q2 Shipment vol.	2021Q2 Market share	Changes every year
Samsung	63	22%	60	18.0%	5.6%
Apple	45	15%	44.4	12%	0.5%
Vivo	25	8.7%	32	10.1%	-22%

Oppo	25	9%	33	11%	-24%
Xiaomi	39	14%	53.2	17%	-25%
Others	90	33	93	30%	-2.6%
Total	287	100%	314	100%	-8.7%

Table 5: worldwide quarterly smartphone shipments, Source: [IDC Corporate., \(2022\)](#)

As per the company, laptops have longer life span as compared to smartphones and this was one of the reasons that earlier, the company stepped back in the mobile industry. Android phones from Samsung, HTC, Motorola, one plus, were over the market and took over the US mobile market with 49% subscribers. Whereas Apple's iPhone is at 32% and Blackberry was at 11%. However, Dell was not a part of this market at that time, so no data was concluded. Dell's smartphones appearance was not domestic. The US based company began selling phones primarily in Brazil, China. The organization has ended the mobile phone market quite a time back, but with the rising demand of the users interests with the laptops and other peripherals, the company might bring back the business with the possibility of increase in the revenue of the company and more customers. [IDC Corporate., \(2022\)](#)

The organization is fully focused on the new project and from the research we found that mobile phone use has become an important part of development efforts, especially in low-income countries. As mobile phone adoption has increased, it has made it easier for the organization to get business out of it. As mobile phones become more prevalent, it will be important to investigate and understand their impact and assess how expanding mobile business will support broader development efforts with the application of

appropriate conceptual frameworks and the rigor of the methodologies adopted. Dell works on Agile and 6 sigma methodology to review every individual and keep every project. On track. The goal of this method is to reduce errors or any defects and use continuous process and project improvement and refine the processes until they produce stable and predictable results as needed by the organization. This result suggests that the impact of this business is divided into three main categories, which is; a purely quantitative approach, rooted in information economics, and a purely qualitative approach, rooted in social implication assessment. Another mixed category of methods provides cheap and timely results but results in less rigor evident at both methodological extremes. This research concludes the investment and launch of a new project in the market. This project will work on the points below as concluded. [Author., Gagandeep Kaur \(2023\)](#)

1. Strategies to bring more revenue from this business.
2. Staying eco-friendly and making sure to go recyclable by the year 2050.
3. Making economical mobile phones for every age group.
4. Developing business in countries like Africa.
5. Delivering the best to the customers and exceeding expectations.
6. Provide sustainable skills development for workers in our supply chains
7. Revolutionize and mutate lives
8. Endorse ethics and privacy
9. Promoting unceasing and green environment

The project's analysis shows that there is significant evidence that mobile phones facilitate access and how mobile phones facilitate new learning and OfCourse the

expansion of the organization. This research includes the way this organization works using different frameworks and methodologies including Agile and 6 sigma to make this project as successful as the other ones. With use of 6 sigma, companies easily focus on reducing defects and errors. This helps to run the projects continuously and improve cycle times while reducing no more than 3.4 defects per million elements. Here the main motive of the company is no doubt to make more money by launching a new project and that too in which the consumer segment is highly interested. As Dell is already into the business of computers and peripherals, so the organization is already aware of the competition in the market. Dell was earlier in this project but stepped back due to the increased demand for computers at that time. But from the learnings and experience, the company is stepping back in again and believes that this time it is going to make an impact and change the perspective of the market and consumers. The approach this company has applied to launch this project is easy to understand and it is an easy step-by-step procedure that the management can follow. As the organization is already a brand name in the market, so, there is no struggle that the company must go through. There is an approach for the plan to be followed and that should be it. But one thing that must keep in mind is that consumers do have their favorite mobile brands already and to break this stereotype is very difficult for the company. This company is working hard now to get back in this business and take over all the loyal computer customers as in for mobiles as well. The simple strategy to get that is to keep the prices low and make sure other developing countries are also considered as a priority under this project. [Author., Gagandeep Kaur \(2023\)](#)

CHAPTER 5. Transformation Project

If we instinctively see organization as more mechanical, possibly we see change as a technological process in which mechanisms are updated or transformed so that they can provide new functionality and language become one of intriguing solutions, constructing blueprints, redistributing features, and amending performance. Then it can be determined what needs to be done, grow a plan to achieve it, and execute the plan to achieve the desired result. Changes in an organization's environment occur over time, and appropriate responses must be made through iterative experimentation and adaptation. Dell Technologies strongly believes in change as change is the only constant in our lives and either its business expansion or including a new change in the company will possibly increase the chances of bringing benefits to the organization. [Author., Gagandeep Kaur \(2023\)](#)

5.1 Expected project benefits

Agile methodologies emphasize continuous face-to-face communication, and communication is cited as the biggest problem in project development. However, Agile methodologies and six sigma strategies have been successfully used in distributed projects, suggesting that they may be useful for new projects and can bring expected benefits for the organization. This study examined the benefits and success of a new project through a global survey. The new generation of smartphones are increasingly seen as portable computers rather than phones due to their powerful on-board computing capabilities, large memory capacity, large screens, and open operating systems that facilitate application development. This research talks about the usage and benefits this

project will bring to the consumers and the organization. Now-a-days most of the people have mobile phones at home rather than landlines and wired phones and so are their children, continuously reliant on mobile phones. [Author., Gagandeep Kaur \(2023\)](#)

1. The company is working to make mobiles pocket friendly. Keeping the prices low so that even the young working generation can afford them easily.
2. Dell chose windows and Android as a platform for these mobile phones, as most of the population uses android or apple these days leading to more consumers.
3. It is an easy way to directly communicate with friends and family on the way and a little device to look for every piece of information with the help of 5G technology which makes it way faster to search for anything and everything.
4. One of the best benefits of launching this business is that mobiles phones can monitor the health and any possible chronic diseases for older people using remote applications. Additionally, the mobile platform can use raw sensor data to identify high-level information based on established medical knowledge, including easily detectable abnormalities such as tachycardia and signs of respiratory infections. Same is the case with other applications, related with banking, online payments, finding locations, emails, staying in touch with loved ones, these are just one touch away. And being Dell is that old customers look upon as a reliable name and go for this mobile brand. [Author., Gagandeep Kaur \(2023\)](#)
5. Dell computers are shipped with built-in applications such as mobile connect. Here one can synchronize their mobile with the laptop and can use a computer just like a mobile phone and vice versa. Using this application with mobile phones, one can easily keep a record of information from the computer and doesn't need to carry

their laptops with them all the time. This will bring immense consumer contribution to the use of these mobile phones. [Author., Gagandeep Kaur \(2023\)](#)

6. New projects bring new job opportunities to everyone. Whenever there is a new project launched by an organization, the opportunity to grow and hire more people comes along with it. Launching mobile phones will give opportunity to people from different segments like mobile industry, software industry, hardware industry to work together and this is helpful for everyone to share new information among each other and provides jobs to everyone. This is also an opportunity for insider employees to grow via the chance of internal job posting. This is a way where companies open new promotions for everyone who can be considered as eligible for this opportunity, leading to an increase in salary and higher position. [Author., Gagandeep Kaur \(2023\)](#)

5.2 Alternative business options

Currently, there are very few people who do not own or use a mobile phone. Today, mobile phones have become an increasing priority in everyone's life, so everyone uses them in their daily lives for communication, business purposes, and other activities. If one looks at it right, mobile phones are an extraordinary invention for mankind, but if one uses them continuously for hours on end and is addicted, then some problems arise. So, along with this mobile phone business, my alternative idea for the organization is to launch new software pre-installed on mobiles named as “Pocket Doctor” by Dell, that can track the heart rate and health of the user while using the phone. Benefits of this application are as below. [Author., Gagandeep Kaur \(2023\)](#)

1. For people of old age, there are many diseases that are beyond the reach of doctors to cure as they are in the last stage or incurable as they were detected when they have already caused the damage to the body and now the chances of curing them are minimal. [Author., Gagandeep Kaur \(2023\)](#)
2. This application will ask the user to register with name, email address, phone number, emergency contact person, full current address details, age, any known diseases, any health issues, known allergies and will be enabled and ON all the time. This will also help the phone user to keep track of his own health. [Author., Gagandeep Kaur \(2023\)](#)
3. One can track one's health every day and can also find out the blood pressure rate, walked steps, pulse rate, heart rate, will ask about eating habits and will track intake of sugar, salt and will enable to notification to the user when it is overconsumed. With the help of Pocket Doctor app, it will be easy for the doctors to find out all the background information of the person's health and any symptoms that can cause the disease. [Author., Gagandeep Kaur \(2023\)](#)
4. If there are any signs of the health dropping down, this application will send a message to the emergency contact number in the app and will alert that person, so that the doctor will be consulted in no time and one's life will be saved. [Author., Gagandeep Kaur \(2023\)](#)
5. This application or software design will be simple to use as keeping in mind that people of old age will be using it, rather than confusing to use, keeping step easy and simple for one to understand. [Author., Gagandeep Kaur \(2023\)](#)

6. In case someone has an old disease, which is going on, this app will also suggest the expert doctors in that terrain. This will help the user to get in touch with that doctor and get more information about their health from the expert. [Author., Gagandeep Kaur \(2023\)](#)
7. This application will also suggest the user suitable diet so that one can stop eating food that can increase chances of increasing problems to one's health. [Author., Gagandeep Kaur \(2023\)](#)
8. As we are focusing on diseases which are threatening for one's life, this app will track every symptom starting from heart rate, blood pressure rate and then the cause of the start of the disease, so that one could have a clear idea where it all went wrong. [Author., Gagandeep Kaur \(2023\)](#)
9. Using this app, one's health will be tracked all the time, keeping one alert and healthy. Just in case there are any emergencies, their loved ones are contacted on time to avoid any mishappening. [Author., Gagandeep Kaur \(2023\)](#)

The benefits of this application are more achievable as it is very informative for everyone. This is a huge help for everyone to stay healthy and find out their current health situation. This app is easy to use and will alert the user when health is not good or there are any severe symptoms tracked. This will surely keep everyone healthy and their loved ones happy. [Author., Gagandeep Kaur \(2023\)](#)

5.3 Expected project disbenefits

Security is a legitimate concern for users of wireless devices and jeopardy caused by these electromagnetic (EM) fields. It is a well-known issue as possible health effects are

caused due to exposure to radio frequency radiation (RFR) especially from mobile phone devices. Signals are transferred via electromagnetic waves through radio frequency and microwave signals which is very harmful for a human body if came in direct interaction. This produces harmful ionizing radiation, which causes health problems especially when we place the phone near our ear skull region which affects our neurons. Below figure shows the effect on our brain.

Nath, A. and Mukherjee, S., (2015)

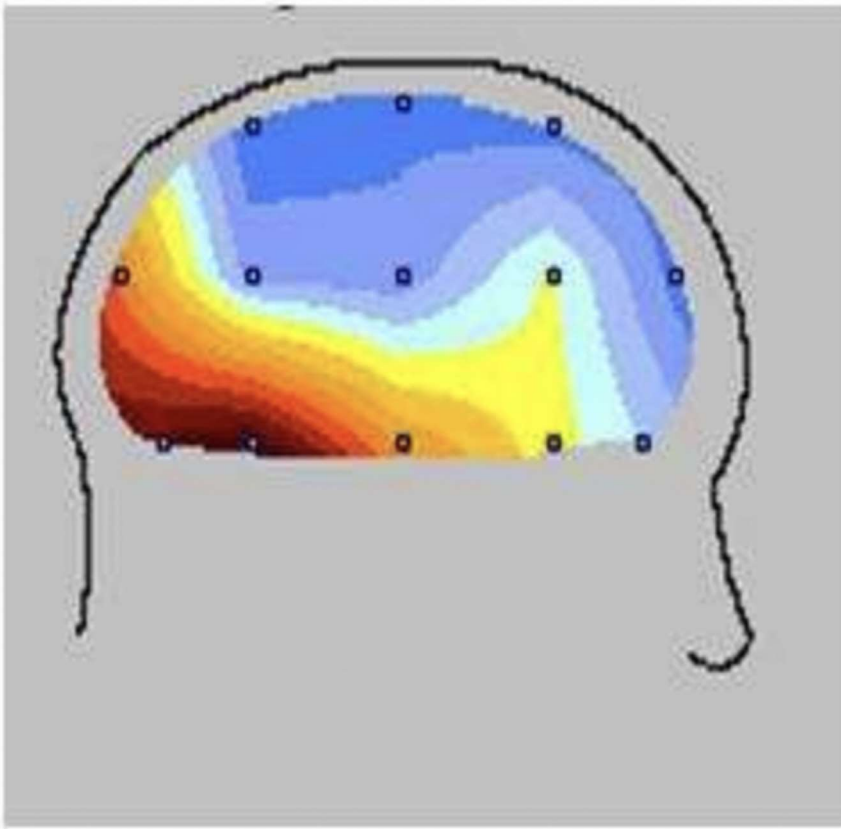


Fig 3: Effect of electromagnetic radiation on human brain; Source: Nath, A. and Mukherjee, S., (2015)

Changing technology and ways of communication has made mobile phones an important part of our lives in this century. Though this device has made our life much easier and communication between our loved ones has become faster, easier and cheaper, especially sitting across in the different parts of the world. We can reach our friends and family in case of any emergency. Everything is just one touch away from us with the use of mobile phones, but despite all the above, mobile phones still cannot eliminate in disadvantages and that will be one of the reasons for the disbenefit of launching this project in the market. But also, one fact that none of the companies consider this as a big reason is that this project might not be a success as it could be harmful for our body because anyhow the consumers and customers will go ahead for the brand and buy it as mobiles have become now the basic need. The major disbenefit of the project can be listed as there are already expert competitors in the market that will give a hard time for the company making an impact on the old users and new ones. Dell, being a brand name, is known for being a well-established computers and peripherals company. As launching mobiles and making sure this project works as wonders is still a challenge here. There are competitors in Android and Apple already and consumers have their favorites already. Dell must make surprisingly good here to make sure this business idea will be successful. As in the past, the company did not make any success in this project, but now the comeback should be stronger than ever. The organization should have some built-in strategies to overcome these challenges. One more thing, Dell is coming up with in partnership with Windows and Android for mobile phones. From the surveys it has been found that consumers don't like windows phones. Tablets and laptops still are preferred with windows over any other platform but customers like Androids or Macintosh platforms

for mobiles. So, bringing this idea of windows phone might not be a good idea for the company. Android has conserved its position as the world's leading mobile operating system in Q4 2022 and is influencing the mobile market with almost 71.8% share. Meanwhile, iOS holds about 27.6% of the mobile operating system markets. This could be one of the reasons that the company might not have a successful place in the mobile market like it has in the computer market. This can give the company a lot of time to adjust and make a name for itself. And when sometimes it takes a lot of time for a company to settle a new project, it might back off causing a loss in investment with time and money. Stakeholders might not invest in similar projects who are already in the market, and it doesn't make any sense to them. They might suggest at the time that this project might not work, and the company should have another idea and can step back as these are some uncertain risks that might happen, and it should not affect their partnership in this project. [Nath, A. and Mukherjee, S., \(2015\)](#)

5.4 Project Approach

As discussed above, this project has been based on agile and 6 sigma methodology. Many companies employ different combinations of manufacturing strategies to improve efficiency, effectiveness, and quality and so is Dell Technologies. The pressure for faster work completion, technical or design complexity has increased the complexity of projects. The project management (PM) literature of the last two decades suggests that the development of PM technology is driven by operational excellence (OE) theory. Agile based Six Sigma methodology reinforces the realization that reducing workflow variability can save both time and money. The integration of Lean Six Sigma (LSS) and Agile Manufacturing (AM) is one of the prominent strategies that has given Dell a competitive

advantage in this modern and competitive era. Lean Six Sigma in Dell is about receiving business knowledge. In a new project or development, finding out a customer's need and implementation, avoiding waste and unimportant tasks, doing everything right the first time and having fun with financiers and developers. With Agile, Dell follows customers' needs and requirements first. My contribution in this research is simply explaining that uncovering customers need is simple and implementing them in modern industry is straight forward. Wiley., 2019

What is Agile: Agile means that planning and execution of the process are in parallel and embracing change as required. All development is time bound – budget constraints take precedence over wish lists, even the essentials of the specification. This keeps neither planning nor engineering requirements obsolete but on the contrary, it is even rigid.

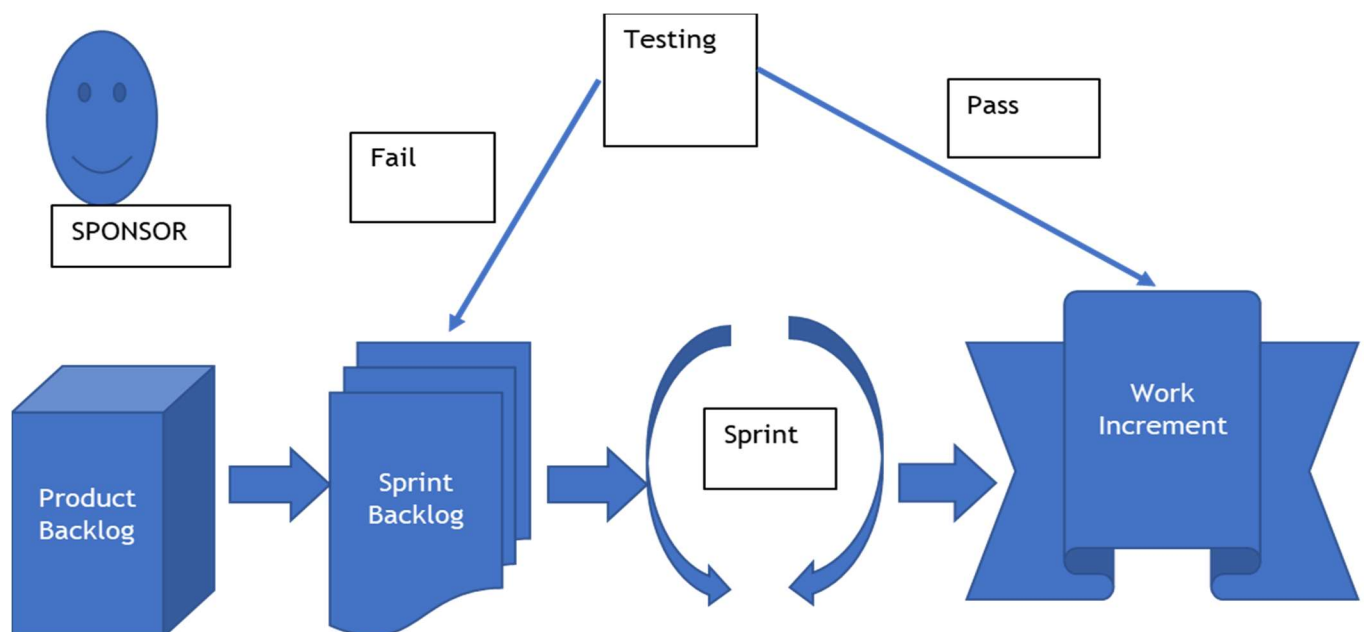


Chart 3: Agile Way for Project; Wiley., 2019

Agile way of planning a project includes below ways:

1. Planning session: To select user stories for the sprint
2. Daily scrum: It includes daily standup meetings which can discuss the achievements, the next plan and any obstacles that are faced.
3. Testing: It includes checking and passing or failing the report. It creates a work system every day in such a way that customers can share feedback immediately.
4. The Sponsor is a person needed for further guidance on the project.

Agile- Good points and curious questions

Embrace Change • Project Planning for mobile phones is not rocket science	Not lean • Sorting user stories and effort estimation in planning poker
• Involves stakeholders • Invests time in PM's experience	When will it finish? • Seeks for what is finished? • Meaning of what means finished?
Includes Knowledge • Gains knowledge as work progresses	Includes priorities • For product backlog and sprint backlog
Communicate • Explain and discuss	Checks if project is a success? • Business goal is reached?

Table 6: Good points with Agile; Author., [Gagandeep Kaur \(2023\)](#)

Six Sigma: A project results should match the plan. Or rather not, ideally, they come alike. Deviations are reduced when the process is well structured, and the goals are known. If in the project, deviation is under control, the outcome of a project becomes predictable.

Six sigma understands and tries to comply with the transfer functions that map the whole project onto expected response and results. [Wiley., 2019](#)

5.5 Proposed project management team structure:

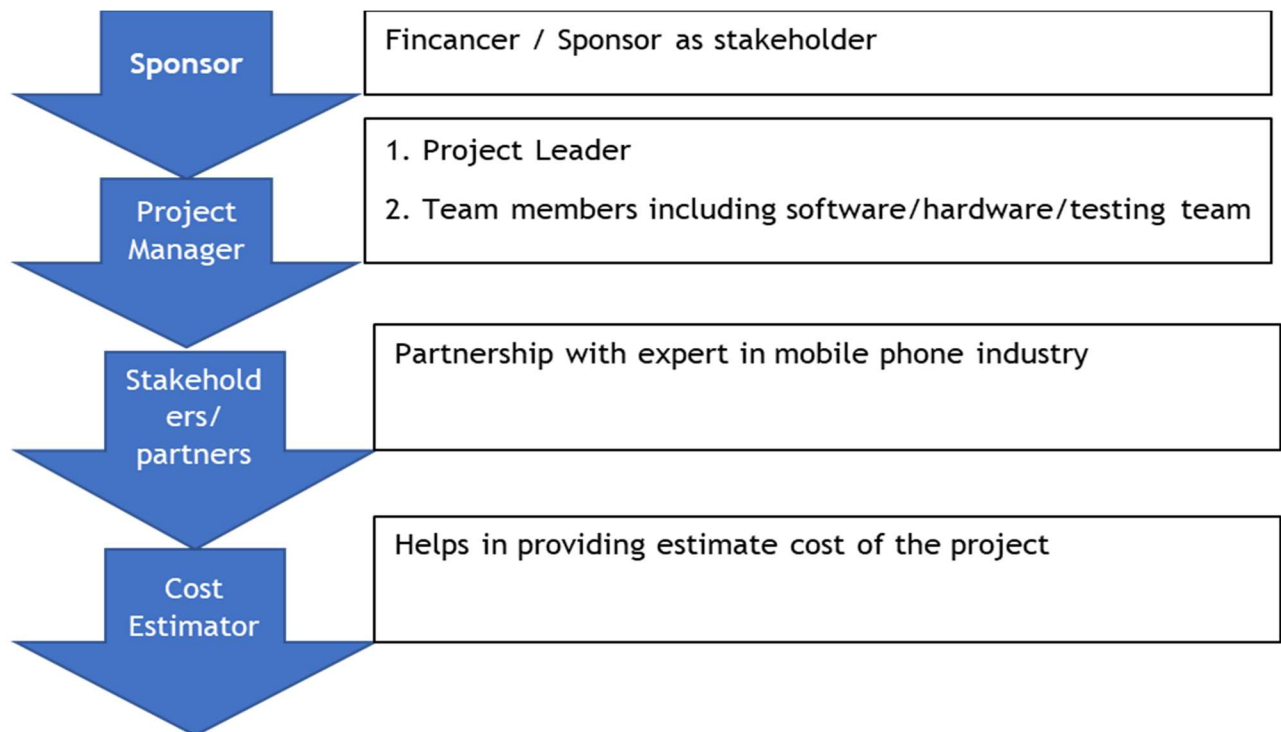


Chart 4: Team Structure; [Author., Gagandeep Kaur \(2023\)](#)

1. Sponsor: This will remain Dell here as the company is already well established and can finance itself for launching a new project of mobile phones in the market.
2. Project Manager: He is responsible for handling a team of specialists who will be working on the project and making sure that the product is tested, and outcome is as required. Project Leader and Team member work under the project manager. Project leader is the supervisor or the specialist who keeps a check on the software, hardware and testing specialists and to guide them where required.

3. Stakeholders/ Partners: This person is the one who is an expert in the designated field of mobile phones to guide Dell when needed.
4. Cost Estimator: Usually gives a clear idea about the estimated investment in this project. **PMBOK,7th edition**

Identification Of critical Roles in line with RACI; **PMBOK,7th edition**

R	Responsible for the project
A	Accountable for making decisions
C	Consult- Stakeholder/Partner for guidance
I	Inform-Needs to know of taken decisions

Table 7: RACI Matrix; **Source: PMBOK,7th edition**

RACI MATRIX						
Taks	Project Sponso r	Project Manager	Project Leader	Team Member s	Stakeholder /Partner	Cost Estimator

Defining Project	I	R	A	I	C	I
Planning Project	R	R	A	I	C	I
Testing	I	C	A	R	I	I
Highlight reports	I	C	A	R	I	I
Making Decisions	A	R	A	I	I	I
Cost Estimation	A	I	I	I	I	R

Table 8: RACI matrix in line with team roles; Source: PMBOK, 7th edition

5.6 Project Plan: The project plan date to launch mobile phones is still not been out fixed but it is expected to be out in a few months.

A Project breakdown structure also known as Work Breakdown Structure (WBS) is a tool you can use for projects, programs, or even initiatives to understand the work required to deliver results. Below is the designed WBS for this project:

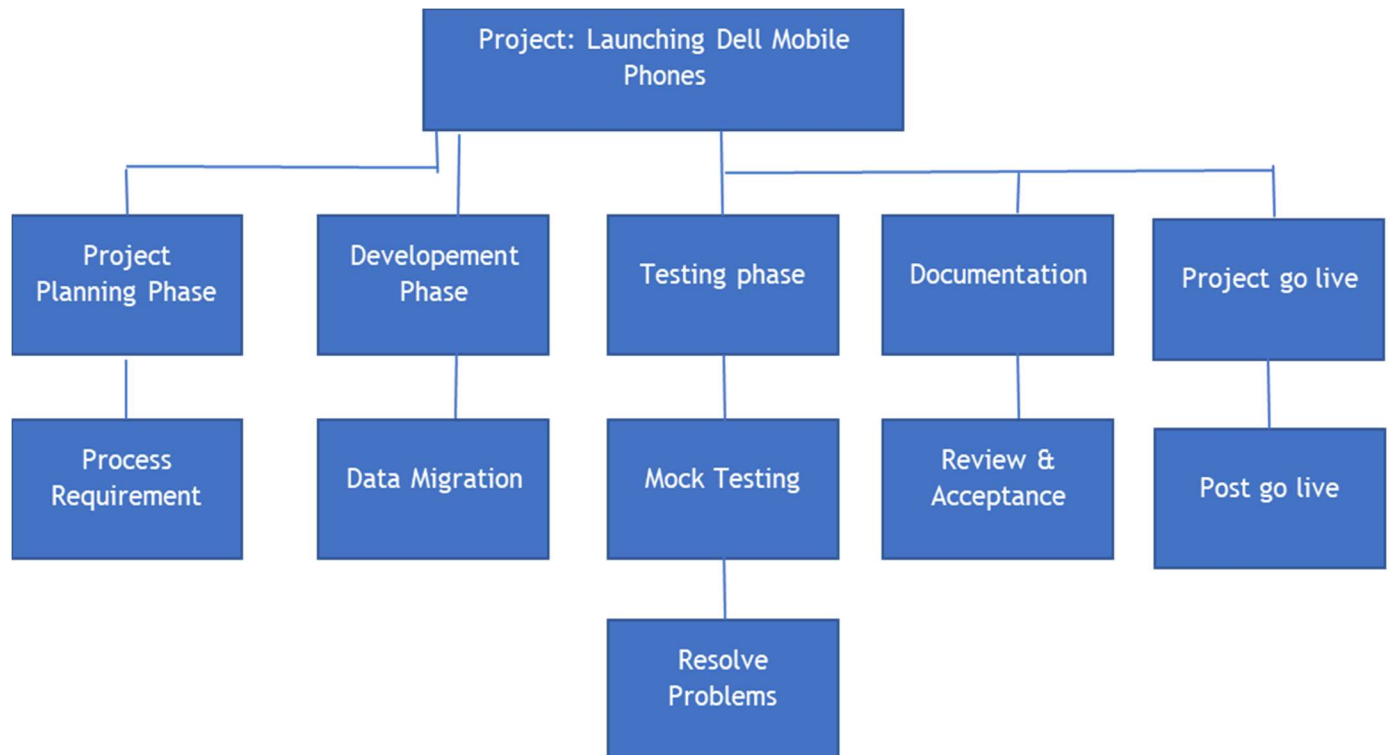


Chart 6: WBS of a Mobile phone project; [Source: Author., Gagandeep Kaur \(2023\)](#)

The WBS of this project is in line with RACI as the person hired for the above designated work is responsible accordingly.

1. In the project planning phase, Project manager and project leader work together to discuss the desired strategy of the project; including the requirements that how it will work and how it should work.
2. The development phase includes budget, the size of the development team and their proficiency level. The need to find capable and skilled people is strongly recommended and here the software/hardware and testing specialists are held responsible.
3. The testing Phase also includes the specialist team checking all the software and hardware components before the product is launched in the market.

4. Documentation is a vital part of any project, and it includes keeping every task on either digital record to make sure that the team can keep a record of what has been done, who has done it and when it has been done. Usually, the Project Leader is responsible for this part.
5. Project go live is the final stage of the project where the product is finally launched and released in the market after all the checkpoints are reviewed and completed.

Author., Gagandeep Kaur (2023)

5.7 Project Cost

Here estimation is time consuming task and usually involves a group of people from different fields which includes a cost estimator who is an expert in the market and gives a brief idea about the market price. For Dell it will include multiple aspects as production cost, retail price, and profit margin also must be kept in mind. Which are in the estimated ratio of 54%, 34% and 12% and could be less or more as these are estimated price lists. So, for this project cost can go as \$500 for production cost, %1099 for retail price and including profit margin, it can vary all the time depending upon the market demand.

DellTechnology., (2022)

5.8 Major Risks in the project

2 major risks that can jeopardize the whole project are already discussed as above:

1. There are already experts in this field who have taken over the mobile market and have been ruling for years. It could be Macintosh operation system or Android, so there is a high chance of consumers denying this product, and which could lead the company towards loss. From research and studies, it shows that Apple has a

huge share of the consumer segment in mobile phones over any brand and it is very difficult to beat this brand in this market. [Apple., \(2023\)](#)

2. Because of the above reason, it can be concluded that the mobile phone project is too ambitious. As scope of the project must be adapted to human needs and financial resources. As possibility of consumers liking this project is very little but still the organization has high hopes and a strong desire to launch the project in the market, even though there is nothing new that will make these mobile phones different from others. [Author., Gagandeep Kaur \(2023\)](#)

5.9 Operational Risks with mitigation plan in the project

Operational risks are the chances of loss attributed to useless or failed inner processes, people, systems, or outside activities that could disrupt the function of commercial enterprise operations. For this project, there are many uncertainties and risks involved in the daily operations and the risks can threaten to reduce productivity, cost and profit for the company. This can include machines not working, power cuts, unnecessary manpower shortage, raw material shortage, management failure. These are some common operational failures that will directly impact the functionality of the project. Simple mitigation plan for this is to keep a backup plan ready 24*7. The company needs to make sure the manpower shortage should not impact the working and should make employees work in different shifts and keep them scheduled as needed with planned leaves and holidays. When any major project is in line, the organizations sometime need to make sure raw material and major supplies must be tracked and inventory needs to be checked on time to make sure they are always in stock. Backups for power cuts also needed to be provided in some areas where this issue is common to avoid any delays in production.

Last but not least, management communication between each other must be followed to avoid any miscommunication over the project that will directly impact all the above discussed points. [Author., Gagandeep Kaur \(2023\)](#)

5.8 Communication Management

Here the organization doesn't need any funding as Dell is a known brand in the market and has billions of dollars of revenue every quarter, but the company has stakeholders in partnership to guide on time for this project who has an expertise in this field. So far Dell has more than 250 global alliances who are most strategic partners including Global systems Integrators, advisories, hyperscale's, cloud service providers, telcos, independent software vendors and many more like VMware, SecureWorks, VitalStream. But for mobile phone project, dell has still not revealed the name of the partner as the company is still in decision process for the same. [DellTechnology., \(2022\)](#)

CHAPTER 6: Reflection

I will complete coursework towards my master's degree this month (Feb 2023). I studied in a British University that has one of the best courses. During this research, there are a few things that I came across that I can happily say that have made my life different than before:

1. My opinion towards the masters' students in our institute underwent a big change. Every teacher who was teaching different modules to students was experienced and an expert in this field. Any student will ask them any random question and they will have answers for each question. OfCourse, all of them has the best knowledge in their field of teaching. Also, how to search and how to make sure that we can easily understand their way of teaching.
2. I also came to realize that a basic level of intelligence is necessary for this research. Even though I am not slow and could easily search for any material online on the internet and from the university's website but still I find that some questions did need more detailed research as some students find it difficult to understand and it took them days to understand how to proceed next, but it could be done. In hours with the right approach and correct guidance.
3. Also, I concluded that studying on your own is also not easy as there are some points when you need help and certainly, I was lucky to discuss issues with teachers and friends to understand a few complicated questions. Completing bachelor's is easy for students but studying masters is not a cup of tea for everyone as it needs detailed research on all the projects that we must work on.

4. Keeping notes and bullet points in one place is a good idea and they are helpful when we do assignments.
5. Trusting your teacher more than yourself is also a learning for me. Many a time, it happened to me, where I would ask a question myself and would have a completely different answer but the approach my teacher would describe to me would make sense then. No doubt the reason is that they are teachers, and we are students.
6. From my masters, I became better at searching now, I can now understand what deep search about any article or journal is. I can easily search any article from an online book now which was a rigid task for me. In the beginning. I remember my first module of Project management tools & Techniques and I would ask several questions to my professor then Dr. Elizabeth George who had immensely loaded patience and would love to explain each question in detail. She was my first teacher to guide me and explained to me how I can do better in my research and kept me self-motivated.
7. I believe I have become a critical thinker; I do find that I have now better analytic and time management abilities. All of which will help me succeed in the workplace. This research has helped me to understand what I am capable of, deepened my ability to carry out any search on any topic and the argument on it and has given me more confidence to develop my skills in making connections between my ideas. This research has helped me in expanding my knowledge in this field.

8. I also learned the importance of prioritization, when to do and what to do and how much time it will take to finish. This made me independent in my thoughts and faster with effective writing and editing skills.
9. Also, one thing that I really loved is working with other students and learned how to finish tasks in groups and how to evaluate and use others knowledge and how I can make use of it with mine. This was an opportunity for me for my professional development.
10. This research will definitely help me in the future with lifelong learning and enhanced professional network with career Advancement and increased earning potential. I have learned how to gain hands in experience in this field and understand the scientific process of developing research questions and testing my hypothesis.
11. Every module I learned new techniques and methods of working in a project and I was not sure if I could be better as I scored good in each module. I would say, I learned from teachers, from the subjects in real time that will help me in my career. From Agile to scrum to different modules that I have studied, have made me think like a Phd student now who can definitely make use of this knowledge in the future career.
12. These above learning will be a part of my future habit now as I am planning to go for my Phd and would love to be an academic researcher or teacher or could be an expert in related field in any Multinational company where an understanding of research would be considered useful. [Author., Gagandeep Kaur \(2023\)](#)

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Dissertation Proposal Form	Gagandeep Kaur	Supervisor Name	Amany Annaggar
Student Name			
Student Number	STU101533	Course	Project Management
Project title:	Launching Dell mobile phones in the market with possibility of increase the revenue of the company and brand name.		
Start and end dates of research:	4th Aug 2022- 4th Aug 2023		
Proposed activity - aims, objectives, research question(s), and state how it is novel			

AIM:

The aim of this project is to launch Dell mobile phones in the market for different individuals in Germany.

Objectives:

- To identify trending mobiles in the telecom industry. Which mobiles are consumers using the most and why?
- To investigate the effectiveness of these mobiles for different individuals in different parts of the country.
- To identify key challenges for users regarding Android users and Windows platform users and find out which one they find easy to use and operate.
- As Dell is a popular brand for laptops and desktops in the market, launching mobiles will increase revenue for the company.
- Making it easy to connect and sync with laptops and desktops so that users can easily communicate with their devices for sharing data.

Research Questions:

1. What could be the benefits of launching Dell Mobiles in the market?
2. What are the concerns users have which affect the full adoption of this brand?
3. What will be the effect of these mobiles on users? Will they accept it or go with other brands they are already using?

Methodology - rationale, data selection and collection, recruitment, participant demographics, analytical process

Methodology

This research will be done through a mixed approach, with quantitative technique being the dominant while also exploiting qualitative approach on the same questionnaires. From the beginning, Dell's strategy was based on many core elements such as build-to-order manufacturing, mass customization, supplier partnerships, just-in-time inventory of components, direct sales, market segmentation, customer service, and more. Sharing data and information with supply partners and customers. But today, Dell and other US personal computer (PC) makers are struggling to make a profit in an environment of falling prices and fierce international competition. We will be using the above strategy for launching mobile phones as well. Secondary data on the other hand is readily available, cheap and saves time, however it is less reliable and maybe outdated.

Rationale

For the past decade, the world has experienced a lot of changes in the telecom industries, especially in mobiles phones. Starting from Landline phone and going to mobiles has bright the world together and made it easy to connect with each other. Now every individual can call and text each other in emergencies. We can send live photos and videos to each other. There are number of advantages that we can talk about these mobile gadgets and launching a Dell mobile has brought its users all these together with a different platform and user interface which will be accepted by everyone.

Therefore, the motive behind this research is to assess the impact mobile has brought upon the end users, evaluate any concerns that users have in order to make informed recommendations that could be helpful to the Individuals, service providers, future investors and the end users at large.

Data selection and Collection -

There are two types of data collection processes, and these are primary data collection and secondary data collection. This research methodology utilises secondary data only.

I will be using my capability on the research with the help of Company's websites, Journals related from University's library. As secondary data is also helpful to understand what previous researchers identified and areas of improvement.

Secondary data collection:

Secondary data is data collected from people other than researchers. In this regard, this study leverages research reports from previous researchers from both existing journals, online sources and the Arden University Digital Library.

Publications from the “Bild” in Germany and other online articles will also be used. Secondary data is important for this study to understand technology trends, discovered challenges, and recommendations from previous researchers. In addition, secondary data is inexpensive and readily available, saving time and complementing primary data.

Participants Demographic - Surveys are distributed along the way from people of all genders and ages, and from different parts of the country, both rural and urban, regardless of education level. This is because everyone has been affected by the advancement of mobile technology and the lack of knowledge in the country which has forced everyone to stay with old phones. Therefore, make sure that the surveys of different people, regardless of group, are consistent.

Analysis of journals, previous scientific studies and publications on the introduction of mobiles will be used as secondary data.

Data Analysis

For analysis, the research data gathered will be presented in forms of graphs, charts, and tabular representation to make it easy to understand. The Microsoft online survey tool provides a summary of all the responses grouped together. This will be efficient, free from human error and easy to use.

The participants will be grouped according to age, level of education, location (rural or urban). The survey results will be summarized and presented in percentage.

Generalizability

You can generalize your findings and use recommendations to improve mobile phone user acceptance. This approach can be applied to other environments as mobile technology is the same across the industry. Therefore, his insights can be used to improve the industry's mobile phones and usage capabilities.

Ethical Issue

There is no request for personal data from participants. You have the option of setting up the survey anonymously so that even researchers cannot trace back to the respondents. Whenever a participant has privacy concerns, they have the option to close the survey immediately.

Literature Review

For years, people had to travel to their loved ones to check on them. However, during the past decade, the world has experienced a major transition from old phones to the smart phones used now a days. With the introduction of Smart phones, tasks in daily lives have become more convenient. The first mobile phone was manufactured by Martin Cooper of Motorola in New York City in 1973 and was equipped with a handset of approximately 2 kilograms in 1979.

Mobile phones are called mobile phones or mobile phones because modern mobile phone services use a cellular network architecture. Mobile phones are considered an important human invention because they are one of the most widely used and sold consumer technology devices. It has grown rapidly in popularity in some areas. In Germany, for example, the total number of mobile phones exceeded the number of households in 1999. Today, mobile phones are widespread around the world, with at least one in almost half of the world's population, with more than 90% of the population owning at least one.

From 1983 to 2014, the number of mobile phone contracts worldwide exceeded 7 billion. Enough to provide one for every human on earth. In the first quarter of 2016, the leading smartphone developers around the world were Samsung, Apple and Huawei. Smartphone sales account for 78% of total mobile phone sales. As of 2016, the best-selling brands of feature phones (slang: "Dumbphone") were Samsung, Nokia and Alcatel. Smartphones have many distinctive features. The International Telecommunication Union measures what is called an active mobile broadband subscription (including tablets) with an internet connection. In developed countries, smartphones have overtaken the use of previous mobile systems. However, in developing countries, they make up about 50% of mobile phones.

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Table: Project timeline and key outputs

Month Activity

November, 2021 Write Brief introduction about project topic and intended outcome

- December 2021 Complete project proposal and ethics Form and send to supervisor.

- Feedback on proposal and ethics form from supervisor
- Review and update the forms.

Submit project proposal and ethics form to ethics committee

- January 2022 Write dissertation Introduction
- Literature Review and research
- Data collection

Data analysis

February 2022 Design solution

Report writing

April 2022 Submission and presentation to supervisor

Supervision Meetings

Date Agenda

22-08-2020 Introductions and discuss research topics

Next Steps

Start on the project proposal: Write research questions, research objectives.

12-11-2021 Discuss next steps; project proposal, ethics form

Next Steps

Draft project proposal, prepare ethics forms

TBA Discuss Proposal and Ethic forms feedback

Next Steps

Update proposal and ethics form for submission

TBA Discuss

TBA TBA

TBA TBA

Research Data Management Plan (Describe the data you expect to acquire or generate during this research project, how you will manage, describe, analyse, and store the data and what mechanisms you will use to share and preserve your data.)

All research data gathered during this project will be stored on Arden One drive anonymously. Online surveys will be collected through a google form, participants are not asked for any personal identifying information. These responses will be downloaded and saved on Arden One drive. All data on Google forms will then be purged. For in-person survey forms, these will be scanned and saved securely on the Arden One drive. All physical copies will be archived safely and await to be destroyed by shredding once the research is complete.

Planned outputs/publications/research datasets/impact/dissemination

When the research is complete, the recommendations shall be published publicly for service providers and users to embrace mobile banking and work on areas of improvement.

If successful, I undertake to carry out the research according to the University's Ethics code of Practice and will be required to complete an Ethics checklist - see relevant forms detailed under Primary Research (Applicant's signature required)	Gagandeep Kaur
Date and signature of Supervisor approval	 <u>06.08.2022</u>
Date and signature of Employer approval (required for Apprentices only)	



Research Ethics Application Form

All research conducted by Arden University students require ethical approval. The application should be sent to your supervisor with your research proposal and any supporting documentation such as a recruitment invitation letter or guide, recruitment flyer (online/offline), participant information sheet, informed consent form, permission letter form an organisation to use their premises, participant instruction guide, questionnaire, measures, interview questions, debrief form, and any supporting or additional documentation that will be provided to the participants, or those helping with the research such as gatekeepers and assistants.

Secondary Research Only (Answer up to and including Question 3)

Complete information up to and including Question 3 if you are conducting ONLY secondary research and can answer YES to the following question:

This investigation will NOT involve the collection of data from human participants, though it may collect data about individuals from published matter (e.g., previously published interviews or behavioural data).

Once the application and research proposal are reviewed and approved by your Supervisor, it will be sent to the Module Leader (ML)/Programme Team Leader (PTL) for final approval. **You cannot collect data until final approval has been provided by the ML/PTL.**

Please complete the information below.

Please circle Yes or No to the following questions and where indicated please provide further information

1. Are you required to use a professional code of ethics and conduct relevant to your profession (e.g. British Psychological Society, Health Care Professions Council National Health Service, Ministry of Defence, the Law Society)

Yes, Arden University Ethics Policy ~~YES/NO~~

If yes, please state

2. If yes, have you read the relevant professional code of ethics and conducts?

YES/~~NO~~

3. Are you sourcing secondary data? (e.g. Information from web sites, journal articles, archives)

YES/~~NO~~

If yes, please provide details

I will be using information related to y research from the required websites, University's journals, Online journals and online information.

(THIS IS THE END OF THE FORM TO BE COMPLETED IF YOU ARE UNDERTAKING SECONDARY RESEARCH)

(IF CONDUCTING PRIMARY RESEARCH, PLEASE COMPLETE THE REST OF THE FORM —— **NO I AM NOT USING PRIMARY DATA**

Primary research is used if your investigation involves the collection of data directly from human participants, rather than depending on data collected from previously completed research, and this can be in the form of a questionnaire, survey, interview or experiments.

All forms for primary research **MUST** accompany this Ethics form, including

A **full** draft of the questionnaire, interview questions, survey or experiment

PLUS:

- Participant Information Sheet
- Participant Informed Consent Form
- Participant Debrief Sheet
- Evidence of organisational approval (where relevant)
- Recruitment poster/invitation letter or email

These forms can be found in the Dissertation Module on ilearn

A proviso is to be used if all of the correct paperwork has been submitted, and this includes a questionnaire that addresses the research aims and objectives but which requires some minor amendments such as phrasing of questions, typos, grammatical errors and so on. Therefore, the form can be approved as long as the conditions of the proviso are met. Guarantee of the changes being met can be authorised by the supervisor and this will save going through the whole proposal/ethics approval process again. The supervisor **MUST** see the proviso changes before the student uses the instrument.

4. Are you using an external research instrument or validated scale? (e.g. survey/psychometric))

YES/NO

If yes, please provide a reference.

5. Are you sourcing primary data involving participants (e.g. Surveys, interviews, focus group, Internet forums)?

YES/NO

6. Are you dealing with sensitive data (e.g. personal data, organisational data, those with vulnerable groups)?

YES/NO

If yes, please outline how this data will be stored securely to ensure compliance with GDPR (all data MUST be stored on the AU Onedrive, not on personal drives)

7. Are you sourcing secondary data? (e.g. Information from web sites, journal articles, archives)

YES/NO

If yes, please provide details

8. Does the study require DBS (Disclosure and Barring Service) checks?

**Please note that unless already working with this population, permission will not be granted to students to data collect from this population)

YES/NO

If yes, please provide serial number, date obtained and expiry data

9. Does the study involve direct contact with:

**Please note that unless already working with this population, permission will not be granted to students to data collect from this population)

Vulnerable adults (e.g. learning difficulties, dementia, living in residential care)
YES/NO

Those under the age of 18 YES/NO

Adults in prison/remanded in custody/on bail
YES/NO

If yes, please outline the participant group

Data Security

10. Can you guarantee the full security and confidentiality of data collected?

YES/NO

If no, please outline reasons

****Please note that any electronic data should be stored on Arden University's one drive (therefore students must use their Arden email address for data collection, communication etc.)**

11. Please outline how you will ensure anonymity and confidentiality of data

12. Will you be responsible for destroying the data after the research is complete?

YES/NO

If yes at what date will the data be destroyed?

If no, who will be responsible?

****Please note that data should be destroyed as soon as possible (when full data usage has been completed) but no later than 3 years from data collection.**

Informed Consent

13. Will all participants receive information as to why the research is being conducted and what their participation will involve?

YES/NO

If no, please state reasons

14. Will all participants be asked to give informed consent before the study starts?

YES/NO

If no, please state reasons

15. Will all participants be told of the data being collected and how the data be used?

YES/NO

If no, please state reasons

16. Will all participants be told that they do not have to participate in the research?

YES/NO

If no, please state reasons

17. Does the study involve deception?

YES/NO

If yes, please provide detail

Risk of Harm

18. Is there any risk that the research may lead to physical/psychological harm or disclosure of criminal activities/convictions?

YES/NO

If yes, please outline and explain what you will do to reduce risks

19. Is there any significant risk that participants may disclose the harming of others or harming of themselves?

YES/NO

If yes, please provide details and actions you will take

Participant Recruitment

20. Are you proposing to recruit participants who are students or staff or Arden University?

YES/NO

If yes, please provide details of any potential conflict of interest and how this will be mitigated

21. Employees of organisations?

YES/NO

If yes, how will permission be gained from the organisation?

22. Students through educational institutions?

YES/NO

If yes, how will permission be gained from the institution?

23. Participants in residential care, social care, nursing homes

YES/NO

If yes, how will permission be gained from the organisation and individuals/carers?

24. Adults in prison, in custody, on remand

YES/NO

If yes, how will permission be gained from the organisation and individuals?

Online Research

25. Will any of your research involve online data collection?

YES/NO

(e.g. online surveys, Facebook, LinkedIn, Twitter)

If yes, how will permission be obtained to collect data if necessary?

26. Will you be using a survey software (e.g. Gorilla or Microsoft Forms)?

YES/NO

If yes, please provide details

Participant Payment

27. Are payments/incentives being offered to participants?

YES/NO

If yes, please provide details

28. Will you tell participants that payment/incentives do not affect participants right to withdraw their data?

YES/NO